Spatial Lag Model Estimation with Sparse Adjustment for Spatial Weight Matrix

Clifford Lam*[1] and Pedro CL Souza†[2]

[1]Department of Statistics, London School of Economics and Political Science
[2]Department of Economics, PUC-Rio

Abstract

Spatial econometric models allow for the effect of spatial interactions among variables through the specification of a spatial weight matrix. While practitioners often face the risk of misspecification of such a matrix, in many problems there are often a number of potential specifications, such as geographic distances or various economic quantities among variables. To facilitate choosing a good spatial weight matrix for modeling, we propose to estimate such by a linear combination of spatial weight matrix specifications, added with a potentially sparse adjustment matrix. The coefficients in the linear combination, together with the sparse adjustment matrix, are subjected to variable selection through the adaptive LASSO. Rate of convergence of all proposed estimators are spelt out when the number of time series variables can grow even faster than the number of time points for data, while oracle properties for all the penalized estimators are presented. Simulations and a real data application demonstrate good performance of our procedure, while highlighting the improvement of the estimated regression coefficients when we include more important spatial weight matrix specifications in our model. Incidentally, our results provide empirical evidence that the choice of spatial weight matrix does matter.

Key words and phrases. Spatial weight matrix; adaptive LASSO; spatial fixed effects; sparse adjustment; partial sign consistency; oracle property.

*Clifford Lam is Associate Professor, Department of Statistics, London School of Economics. Email: C.Lam2@lse.ac.uk

†Pedro CL Souza is Assistant Professor, Department of Economics, PUC-Rio. Email: pedroclsouza@econ.puc-rio.br

1 Introduction

Spatial econometrics include spatial interactions in the study of various variables through specific econometric models. Such interactions come in many forms. Examples include the effects of learning in classroom in social study (Ammermuller and Pischke, 2009, Angrist and Lang, 2004), the spreading of crime and delinquent behavior (Glaeser et al., 1996) in criminology, disease mapping in geographical epidemiology and the contagion between financial markets (Longstaff, 2010). In spatial econometric models, such interactions are usually assumed known and are given through the so-called spatial weight matrix, which is a square matrix of size N with N being the number of time series variables we are modeling.

There are several potential problems applied researchers may face when using these models. On the one hand, there may not be enough expert knowledge in specifying an $N \times N$ spatial weight matrix, especially when N is large relative to the sample size T, since the number of specifications has the order of N^2. To overcome this problem, a few researchers have proposed to estimate the spatial weight matrix on top of other model parameters using data available. See for example, Pinkse et al. (2002), Beenstock and Felsenstein (2012), Bhattacharjee and Jensen-Butler (2013) or Lam and Souza (2015a).

On the other hand, a researcher may have too many potential specifications using some pre-specified criteria in constructing the spatial weight matrix. For instance, using geographical distance, one can specify w_{ij} in the spatial weight matrix $\mathbf{W}$ by calculating the inverse of the distance r^{-1} between units i and j. One can in fact use $r^{-\ell}$ with ℓ being a positive number to specify a spatial weight matrix. A simple adjacency matrix can also play the role of a spatial weight matrix. Other examples include the relative amount of import or export or relative GDP over a region.

One solution is to allow for multiple spatial weight matrices in the model. In Arnold et al. (2011), three spatial weight matrices are specified for modeling the stock returns of the Euro Stoxx 50 members, constructed using the relative market capitalization, adjacency by industrial branches, and adjacency by country of origin respectively. Then a linear combination of them is used with the coefficients of such linear combination estimated through data using generalized method of moments. While Arnold et al. (2011) assumed a diagonal covariance matrix for the model errors and normality with serial independence, they also have a simple model with any regressors.

In this paper, we generalize the model in Arnold et al. (2011) to include both exogenous or endogenous regressors and fixed effects, while allowing for non-Gaussian errors with more general serial dependence and covariance structure. The major generalization, however, is to allow for a sparse adjustment matrix to be added to the linear combination of spatial weight matrix specifications. This allows for certain major departure of spatial interactions from what the best linear combination can achieve, especially when all the specified spatial weight matrices may not be quite capturing the spatial interactions among all N variables. Such a sparse adjustment to the linear combination is also estimated from the data. Instead of using quasi-likelihood or generalized method of moments like the FGMM in Fan and Liao (2014), we use the adaptive LASSO to penalize the elements in the sparse adjustment matrix, and also use it to select which spatial weight matrix specifications are relevant in a linear combination of them. Like Fan and Liao (2014), we also assume that there are exogenous covariates available, which is then used similar to instrumental variables in uncoupling the intrinsic endogeneity of the observed variables in a spatial lag model. In a different context and method, sparsity and exogenous variables are also assumed in Ahrens and Bhattacharjee (2015), for instance.

We prove that the zeros in the sparse adjustment matrix and also the zeros in the coefficients of the spatial weight matrix linear combination can be selected consistently, and also the asymptotic normality for the non-zero coefficients. We are also able to prove the asymptotic normality of the regression coefficient estimators in the spatial lag model. These results mean that we can carry out inference not only on the regression parameters, but also the non-zero entries in the sparse adjustment matrix, meaning that we can test how much individual spatial interactions are away from the "best" linear combination of spatial weight matrices. We can also test how relevant each spatial weight matrix specification is by looking at the estimated coefficients for the linear combination of spatial weight matrices. In Section 5.2, we demonstrate that including more "relevant" spatial weight matrix specifications improves the significance of important regressors in the model. Incidentally, it shows that a better specification of spatial weight matrix can be important to the estimation of model parameters.

The rest of the paper is organized as follows. Section 2 presents the spatial lag model and the sparse adjustment idea on a linear combination of spatial weight matrices. The LASSO and adaptive LASSO estimation problems for various parameters in the model are also presented. Section 3 presents all the assumptions in the paper, as well the the sign consistency and asymptotic normality of our estimators, while identification of parameters are done asymptotically. Section 4 presents the algorithm used for calculating our estimators, and also the BIC criterion for finding suitable tuning parameters. A two-stage least squares extension to our method is also discussed. Section 5 provides detailed simulations to demonstrate finite sample performance of our estimators, while Section 5.2 provides a thorough empirical study on how shocks in a nation's Gross Domestic Product propagate to neighboring countries. Proofs of all theorems are given in the supplementary materials for this paper.

2 Model and Methodology

We consider the following spatial lag model with fixed effects,

$$\mathbf{y}_t = \boldsymbol{\mu}^* + \mathbf{W}^*\mathbf{y}_t + \mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, \quad t = 1, \dots, T, \tag{2.1}$$

where $\mathbf{y}_t = (y_{t1}, \dots, y_{tN})^{\mathrm{T}}$ is an $N \times 1$ vector of dependent time series variables. The matrix $\mathbf{W}^*$ is an $N \times N$ spatial weight matrix with 0 on the main diagonal. It has possibly negative off-diagonal elements, and is not necessarily symmetric. These allow for both positive and negative, and possibly asymmetric spatial interactions among the component time series. The vector $\boldsymbol{\mu}^*$ is an $N \times 1$ constant vector, while the spatial regression parameter $\boldsymbol{\beta}^*$ has size $K \times 1$, so that the matrix of covariates $\mathbf{X}_t$ has size $N \times K$. The innovation process $\{\boldsymbol{\epsilon}_t\}$ has mean $\mathbf{0}$ and covariance matrix $\boldsymbol{\Sigma}_\epsilon$. For more detailed assumptions, see Section 3.1.

In spatial econometrics literature, the matrix $\mathbf{W}^*$ is usually assumed to be known (up to an unknown multiplicative constant ρ^*, called the spatial correlation). As outlined in the introduction, there were many attempts to estimate the spatial weight matrix from data, avoiding completely its specification. In this paper, we take a different approach. We make use of possible expert knowledge about $\mathbf{W}^*$, while allowing the final estimator to deviate slightly from it. More specifically, we decompose $\mathbf{W}^* = \mathbf{A}^* + \rho^*\mathbf{W}_0$, where $\mathbf{W}_0$ is the pre-specified spatial weight matrix, and $-1 \leq \rho^* \leq 1$ is the spatial correlation of the model, which adjusts the magnitude and direction of the spatial interactions specified in $\mathbf{W}_0$. If $\mathbf{W}_0$ exactly represents the true underlying spatial interaction pattern, then $\mathbf{A}^* = \mathbf{0}$. If $\mathbf{W}_0$ is close to the true underlying spatial interaction pattern, then $\mathbf{A}^*$ may not be exactly $\mathbf{0}$, but should be expected to be sparse or approximately sparse (i.e., with a lot of "small" elements). Hence, $\mathbf{A}^*$ can be interpreted as a sparse adjustment to the matrix $\mathbf{W}_0$, and model (2.1) becomes

$$\mathbf{y}_t = \boldsymbol{\mu}^* + (\mathbf{A}^* + \rho^*\mathbf{W}_0)\mathbf{y}_t + \mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, \quad t = 1, \dots, T. \tag{2.2}$$

Note that the diagonal elements of $\mathbf{A}^*$ are all 0.

As a further generalization which can be of practical use to applied researchers, we assume that there can be more than one potential specification of the spatial weight matrix. This is particularly true if a researcher can specify the matrix using different criteria. For instance, using the inverse of geographical distance, cubic inverse of the distance, and some measurements of economic ties between two countries can produce 3 rather different specifications. The researcher then faces the problem of which specification to use. With sparse adjustment in mind, we propose to decompose the true spatial weight matrix $\mathbf{W}^*$ into the following:

$$\mathbf{W}^* = \mathbf{A}^* + \sum_{i=1}^{M} \delta_i^* \mathbf{W}_{0i}, \quad -1 \leq \rho^* = \sum_{i=1}^{M} \delta_i^* \leq 1. \tag{2.3}$$

In the above, $\mathbf{A}^*$ is the sparse adjustment described before. The spatial correlation ρ^* is now the sum of the δ_i^*'s when $\mathbf{A}^* = \mathbf{0}$. It can be considered a generalization of the traditional definition of spatial correlation if $\mathbf{A}^* \neq \mathbf{0}$. The specification $\mathbf{W}_0$ described before is now replaced by a linear combination of potential specifications $\mathbf{W}_{0i}$, $i = 1, \dots, M$, so that M denotes the total number of specifications. This way we can penalize the δ_i's in model (2.8) below to see which specification, or a linear combination of them, is the best suited for the data. As can be seen in the real data analysis in Section 5.2, including important specifications in the model can improve the estimation of the spatial regression coefficients significantly, and our procedure in penalizing the δ_i's facilitates this.

With sparse adjustment to a linear combination of specified spatial weight matrices, the complete model then reads

$$\mathbf{y}_t = \boldsymbol{\mu}^* + \left(\mathbf{A}^* + \sum_{i=1}^{M} \delta_i^* \mathbf{W}_{0i}\right)\mathbf{y}_t + \mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, \quad t = 1, \dots, T. \tag{2.4}$$

With $\boldsymbol{\mu}^*$, the spatial fixed effects of the model is then $(\mathbf{I}_N - \mathbf{W}^*)^{-1}\boldsymbol{\mu}^*$. For identifiability of such, we assume without loss of generality that $E(X_t) = \mathbf{0}$. If not, we can write

$$\mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\mu}^* = (\mathbf{X}_t - E(\mathbf{X}_t))\boldsymbol{\beta}^* + (\boldsymbol{\mu}^* + E(\mathbf{X}_t)\boldsymbol{\beta}^*),$$

so that the spatial fixed effects is now captured by $\boldsymbol{\mu}^* + E(\mathbf{X}_t)\boldsymbol{\beta}^*$ rather than $\boldsymbol{\mu}^*$, and the covariates are of mean $\mathbf{0}$.

All the above motivate us to consider penalized estimation of $\mathbf{A}^*$ and the δ_i's. For the rest of the paper, we focus on analyzing model (2.4), since model (2.2) is in fact a special case of (2.4). We rewrite (2.4) as

$$\mathbf{y} = \boldsymbol{\mu}^* \otimes \mathbf{1}_T + \mathbf{Z}\boldsymbol{\xi}^* + \mathbf{Z}\mathbf{V}_0\boldsymbol{\delta}^* + \mathbf{X}_{\boldsymbol{\beta}^*}\mathrm{vec}(\mathbf{I}_N) + \boldsymbol{\epsilon}, \tag{2.5}$$

where $\mathbf{y} = \mathrm{vec}\{(\mathbf{y}_1, \dots, \mathbf{y}_T)^{\mathrm{T}}\}$, $\boldsymbol{\epsilon} = \mathrm{vec}\{(\boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_T)^{\mathrm{T}}\}$, $\mathbf{Z} = \mathbf{I}_N \otimes (\mathbf{y}_1, \dots, \mathbf{y}_T)^{\mathrm{T}}$, $\boldsymbol{\delta}^* = (\delta_1^*, \dots, \delta_M^*)^{\mathrm{T}}$, $\mathbf{X}_{\boldsymbol{\beta}^*} = \mathbf{I}_N \otimes \{(\mathbf{I}_T \otimes \boldsymbol{\beta}^{*\mathrm{T}})(\mathbf{X}_1, \dots, \mathbf{X}_T)^{\mathrm{T}}\}$, $\boldsymbol{\xi}^* = \mathrm{vec}(\mathbf{A}^{*\mathrm{T}})$, and $\mathbf{V}_0 = (\mathrm{vec}(\mathbf{W}_{01}^{\mathrm{T}}), \dots, \mathrm{vec}(\mathbf{W}_{0M}^{\mathrm{T}}))$. The notation $\otimes$ represents the Kronecker product, and $\mathbf{1}_m$ is a vector of ones of size m.

This model treats $\mathbf{Z}$ and $\mathbf{Z}\mathbf{V}_0$ as design matrices in a classical linear regression setting with $\boldsymbol{\xi}^*$ and $\boldsymbol{\delta}^*$ the regression parameters, except for the fact that $\mathbf{Z}$ contains the endogenous variables $\mathbf{y}_t$. In view of the endogeneity of $\mathbf{Z}$, in this paper we assume that exogenous variables $\mathbf{B}_t$ for $t = 1, \dots, T$, each of size $N \times L$ with $L \geq K$, are available to us. It means that $\mathbf{B}_t$ is independent of $\boldsymbol{\epsilon}_t$ for each t, but is correlated with $\mathbf{X}_t$ in general. Hence, $\mathbf{B}_t$ serves a function similar to an instrument for model (2.5), since it only correlates with $\mathbf{y}_t$ and $\mathbf{X}_t$ but not $\boldsymbol{\epsilon}_t$. If $\mathbf{X}_t$ is exogenous, then we can take $\mathbf{B}_t = \mathbf{X}_t$. To utilize $\{\mathbf{B}_t\}$, define $\bar{\mathbf{B}} = T^{-1}\sum_{t=1}^{T}\mathbf{B}_t$, $\boldsymbol{\gamma} = L^{-1}\mathbf{1}_L$, and

$$\mathbf{B} = T^{-1/2}N^{-a/2}(\mathbf{B}_{\boldsymbol{\gamma}} - \bar{\mathbf{B}}_{\boldsymbol{\gamma}}) = T^{-1/2}N^{-a/2}\mathbf{I}_N \otimes \{(\mathbf{I}_T \otimes \boldsymbol{\gamma}^{\mathrm{T}})(\mathbf{B}_1 - \bar{\mathbf{B}}, \dots, \mathbf{B}_T - \bar{\mathbf{B}})^{\mathrm{T}}\}. \tag{2.6}$$

We assume that $\mathbf{B}_t$ varies over time, such that $\mathbf{B}_t \neq \bar{\mathbf{B}}$ for at least one $t \in \{1, \dots, T\}$. The constant $a \in [0, 1]$ is defined in Assumption R4 in Section 6, and in general a larger a means the exogenous variables in $\mathbf{B}_t$ are correlated with more covariates in $\mathbf{X}_t$. In practice, we do not know a, and $\mathbf{B}$ is calculated by dividing by $N^{1/2}$. This affects only the optimal values of the tuning parameters γ_T and γ_T' in (2.8) and (2.11) respectively. See Section 4 for practical details in finding the best tuning parameters.

The vector $\boldsymbol{\gamma}$ defined above is a simple way to aggregate the exogenous variables. However, this can actually be estimated from data as well. See the two-stage least squares extension described in Section 4.1.

With $\mathbf{B}$ defined in (2.6), we can define the augmented model as

$$\mathbf{B}^{\mathrm{T}}\mathbf{y} = \mathbf{B}^{\mathrm{T}}\mathbf{Z}(\boldsymbol{\xi}^* + \mathbf{V}_0\boldsymbol{\delta}^*) + \mathbf{B}^{\mathrm{T}}\mathbf{X}_{\boldsymbol{\beta}^*}\mathrm{vec}(\mathbf{I}_N) + \mathbf{B}^{\mathrm{T}}\boldsymbol{\epsilon}. \tag{2.7}$$

One motivation for this definition is that the spatial fixed effects are gone since $\mathbf{B}^{\mathrm{T}}(\boldsymbol{\mu}^* \otimes \mathbf{1}_T) = \mathbf{0}$. Another motivation is that the correlation between the "response" $\mathbf{B}^{\mathrm{T}}\mathbf{y}$ and the "error" $\mathbf{B}^{\mathrm{T}}\boldsymbol{\epsilon}$ is now weakened because of the independence between $\mathbf{B}$ and $\boldsymbol{\epsilon}$. This means that consistent estimation for $\boldsymbol{\xi}^*$ and $\boldsymbol{\delta}^*$ are possible again. This augmented model means that we are modeling

$$\widetilde{\mathbf{y}}_j = T^{-1/2}N^{-a/2}\sum_{t=1}^{T}\boldsymbol{\gamma}^{\mathrm{T}}(\mathbf{b}_{t,j} - \bar{\mathbf{b}}_j)\mathbf{y}_t, \quad j = 1, \ldots, N,$$

where $\mathbf{b}_{tj}^{\mathrm{T}}$ is the j-th row of $\mathbf{B}_t$, and $\bar{\mathbf{b}}_j$ is the sample mean of $\{\mathbf{b}_{t,j}\}_{1\le t\le T}$.

For the penalized estimation, similar to Lam and Souza (2015b), we propose a LASSO penalization problem to estimate $\boldsymbol{\theta}^* = (\boldsymbol{\xi}^{*\mathrm{T}}, \boldsymbol{\delta}^{*\mathrm{T}})^{\mathrm{T}}$ by $\widetilde{\boldsymbol{\theta}}$:

$$\begin{aligned}
\widetilde{\boldsymbol{\theta}} &= \arg\min_{\boldsymbol{\theta}} \frac{1}{2T}\big\|\mathbf{B}^{\mathrm{T}}\mathbf{y} - \mathbf{B}^{\mathrm{T}}\mathbf{Z}\boldsymbol{\xi} - \mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0\boldsymbol{\delta} - \mathbf{B}^{\mathrm{T}}\mathbf{X}_{\boldsymbol{\beta}}\mathrm{vec}(\mathbf{I}_N)\big\|^2 + \gamma_T\big\|\boldsymbol{\xi}\big\|_1, \\
&\quad \text{subj. to } \sum_{j=1}^{N}\Big|a_{ij} + \sum_{r=1}^{M}\delta_r w_{0,r,ij}\Big| < 1, \text{ with } -1 \le \sum_{i=1}^{M}\delta_i \le 1, \\
\boldsymbol{\beta} &= \boldsymbol{\beta}(\boldsymbol{\theta}) = (\mathbf{X}^{\mathrm{T}}\mathbf{B}^v\mathbf{B}^{v\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{B}^v\mathbf{B}^{v\mathrm{T}}\Big(\mathbf{I}_{TN} - \mathbf{A}^{\otimes} - \sum_{i=1}^{M}\delta_i\mathbf{W}_{0i}^{\otimes}\Big)\mathbf{y}^v,
\end{aligned} \tag{2.8}$$

where we denote a_{ij} the (i,j)th entry of $\mathbf{A}$ and $w_{0,r,ij}$ the (i,j)th entry of $\mathbf{W}_{0r}$. The definitions of $\boldsymbol{\xi}$, $\boldsymbol{\delta}$ and $\boldsymbol{\beta}$ are parallel to those in (2.5). Also, we define $\mathbf{y}^v = (\mathbf{y}_1^{\mathrm{T}}, \ldots, \mathbf{y}_T^{\mathrm{T}})^{\mathrm{T}}$, $\mathbf{B}^v = ((\mathbf{B}_1 - \bar{\mathbf{B}})^{\mathrm{T}}, \ldots, (\mathbf{B}_T - \bar{\mathbf{B}})^{\mathrm{T}})^{\mathrm{T}}$, $\mathbf{X} = (\mathbf{X}_1^{\mathrm{T}}, \ldots, \mathbf{X}_T^{\mathrm{T}})^{\mathrm{T}}$, and for any matrix M, $M^{\otimes} = \mathbf{I}_T \otimes M$. The above penalizes on the L_1 norm of the vector $\boldsymbol{\xi}$, defined by $\big\|\boldsymbol{\xi}\big\|_1 = \sum_j |\xi_j|$, which does not involve the δ_j's. We do not penalize $\boldsymbol{\delta}$ at this stage since the analysis of the theoretical properties of $\widetilde{\boldsymbol{\xi}}$ is easier without penalizing both $\boldsymbol{\xi}$ and $\boldsymbol{\delta}$ at the same time.

The choice of the function $\boldsymbol{\beta}(\cdot)$ in the definition (2.8) is motivated as follows. Consider stacking the response vectors $\mathbf{y}_t$ into $\mathbf{y}^v = (\mathbf{y}_1^{\mathrm{T}}, \ldots, \mathbf{y}_T^{\mathrm{T}})^{\mathrm{T}}$, so that model (2.4) becomes

$$\mathbf{y}^v = \mathbf{1}_T \otimes \boldsymbol{\mu}^* + \Big(\mathbf{A}^{*\otimes} + \sum_{i=1}^{M}\delta_i^*\mathbf{W}_{0i}^{\otimes}\Big)\mathbf{y}^v + \mathbf{X}\boldsymbol{\beta}^* + \boldsymbol{\epsilon}^v,$$

where $\boldsymbol{\epsilon}^v$ is defined similar to $\mathbf{y}^v$. Multiplying both sides by $\mathbf{B}^{v\mathrm{T}}$, since $\mathbf{B}^{v\mathrm{T}}(\mathbf{1}_T \otimes \boldsymbol{\mu}^*) = \mathbf{0}$, we have

$$\mathbf{B}^{v\mathrm{T}}\Big(\mathbf{I}_{TN} - \mathbf{A}^{*\otimes} - \sum_{i=1}^{M} \delta_i^* \mathbf{W}_{0i}^{\otimes}\Big)\mathbf{y}^v = \mathbf{B}^{v\mathrm{T}}\mathbf{X}\boldsymbol{\beta}^* + \mathbf{B}^{v\mathrm{T}}\boldsymbol{\epsilon}^v.$$

The least square estimator for $\boldsymbol{\beta}^*$ from the above will be $\boldsymbol{\beta}(\boldsymbol{\theta})$ as defined in (2.8).

We do not choose to use the maximum likelihood approach in our setting since we do not assume normality of the residuals. In fact the residuals are allowed to have serial dependence, as defined in Section 3.1. Comparing to the QMLE approach from Lee and Yu (2010), the need that both T and N diverge at the same time in our paper arise from the fact that the spatial weight matrices are parameters to be estimated rather than fixed as in Lee and Yu (2010), which increases the difficulty substantially since the vector of unknown is of order N^2. Still there are settings where N can be growing faster than T in our paper; see Remark 2 in Section 3.4. This also explains why we do not use the generalized method of moments (GMM) because it is also difficult to implement under such a scenario. A penalized QMLE or GMM approach could be possible, and we leave it for future research.

In Section 4, we present a modified block coordinate descent (BCD) algorithm which is adapted to solving (2.8). The idea is similar to the one introduced in Lam and Souza (2015b) for solving a different LASSO minimization. With $\widetilde{\boldsymbol{\theta}}$, we then have the LASSO estimator $\widetilde{\boldsymbol{\beta}} = \boldsymbol{\beta}(\widetilde{\boldsymbol{\theta}})$, which is taken as the estimator of $\boldsymbol{\beta}^*$. The tuning parameter γ_T will be specified in Section 3.1 when we introduce various assumptions for our theoretical results. The explicit formula for the estimator $\widetilde{\boldsymbol{\delta}}$, which is a profile least square type of estimator by profiling out $\boldsymbol{\beta}$ using the special form defined in (2.8), is given by

$$\begin{aligned}
\widetilde{\boldsymbol{\delta}} &= \Big[(\mathbf{H} - \mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0)^{\mathrm{T}}(\mathbf{H} - \mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0)\Big]^{-1}(\mathbf{H} - \mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0)^{\mathrm{T}} \\
&\quad \cdot (\mathbf{B}^{\mathrm{T}}\mathbf{Z}\widetilde{\boldsymbol{\xi}} - \mathbf{B}^{\mathrm{T}}\mathbf{y} + \mathbf{K}(\mathbf{I}_{TN} - \widetilde{\mathbf{A}}^{\otimes})\mathbf{y}^v), \text{ with} \\
\mathbf{K} &= T^{-1/2}N^{-a/2}\Big(\sum_{t=1}^{T} \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}})\boldsymbol{\gamma}\Big)(\mathbf{X}^{\mathrm{T}}\mathbf{B}^v\mathbf{B}^{v\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{B}^v\mathbf{B}^{v\mathrm{T}}, \\
\mathbf{H} &= \mathbf{K}(\mathbf{W}_{01}^{\otimes} \cdots \mathbf{W}_{0M}^{\otimes})(\mathbf{I}_M \otimes \mathbf{y}^v).
\end{aligned} \tag{2.9}$$

In practice, we do not need to specify a value of a in the definition of $\mathbf{K}$, since $N^{a/2}$ is canceled out in the formula for $\widetilde{\boldsymbol{\delta}}$ above. To achieve partial sign consistency - to be presented in details in the context of this paper in Section 3.4 - of the estimator for $\mathbf{A}^*$, we need to improve the tuning parameter γ_T to adapt to different magnitudes of the ξ_j's. This is because for the LASSO estimator to be sign consistent, technical conditions like the irrepresentable condition in Zhao and Yu (2006) has to be satisfied, which is difficult to achieve without stringent assumptions.

This calls for the following adaptive LASSO problem (see Zou (2006) for more details):

$$
\begin{aligned}
\widehat{\boldsymbol{\xi}} &= \arg\min_{\boldsymbol{\xi}} \frac{1}{2T}\big\|\mathbf{B}^{\mathrm{T}}\mathbf{y} - \mathbf{B}^{\mathrm{T}}\mathbf{Z}\boldsymbol{\xi} - \mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0\widetilde{\boldsymbol{\delta}} - \mathbf{B}^{\mathrm{T}}\mathbf{X}_{\widetilde{\boldsymbol{\beta}}}\mathrm{vec}(\mathbf{I}_N)\big\|^2 + \gamma_T\mathbf{v}^{\mathrm{T}}|\boldsymbol{\xi}|,\\
&\text{subj. to } \sum_{j=1}^{N}\Big|a_{ij} + \sum_{r=1}^{M}\widetilde{\delta}_r w_{0,r,ij}\Big| < 1,
\end{aligned}
\tag{2.10}
$$

where $\mathbf{v} = (|\widetilde{\xi}_1|^{-1}, \ldots, |\widetilde{\xi}_{N^2}|^{-1}|)^{\mathrm{T}}$, $|\boldsymbol{\xi}| = (|\xi_1|, \ldots, |\xi_{N^2}|)^{\mathrm{T}}$. We replace the variables $\boldsymbol{\beta}$ and $\boldsymbol{\delta}$ by their respective LASSO estimators $\widetilde{\boldsymbol{\beta}}$ and $\widetilde{\boldsymbol{\delta}}$, so that the above becomes a proper adaptive LASSO problem, which can be solved by one more step of the modified BCD algorithm to be introduced in Section 4. The tuning parameter γ_T stays the same as in the LASSO problem (2.8). The weight associated with ξ_j is now $\gamma_T/|\widetilde{\xi}_j|$, so that if $\widetilde{\xi}_j$ is close enough to ξ_j^*, then the penalization is more scalable to each ξ_j. For instance, if $\xi_j^* = 0$ and $\widetilde{\xi}_j$ is close to 0, then the weight associated with ξ_j above is $\gamma_T/|\widetilde{\xi}_j|$ which is large, hence ξ_j should be penalized to 0, the correct magnitude, by this large penalization weight. In Theorem 4, we present the asymptotic normality of $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}(\widehat{\boldsymbol{\xi}}, \widetilde{\boldsymbol{\delta}})$, so that the adaptive LASSO estimator $\widehat{\boldsymbol{\xi}}$ is important in the inference of the regression parameters $\boldsymbol{\beta}^*$ in the model.

We need to penalize on the elements in $\boldsymbol{\delta}$ as well, since we want to select which $\mathbf{W}_{0r}$, $r = 1, \ldots, M$, or a linear combination of them, best estimate the true spatial weight matrix. To this end, we propose the following:

$$
\begin{aligned}
\widehat{\boldsymbol{\delta}} &= \arg\min_{\boldsymbol{\delta}} \frac{1}{2T}\big\|\mathbf{B}^{\mathrm{T}}\mathbf{y} - \mathbf{B}^{\mathrm{T}}\mathbf{Z}\widehat{\boldsymbol{\xi}} - \mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0\boldsymbol{\delta} - \mathbf{B}^{\mathrm{T}}\mathbf{X}_{\boldsymbol{\beta}(\widehat{\boldsymbol{\xi}},\boldsymbol{\delta})}\mathrm{vec}(\mathbf{I}_N)\big\|^2 + \gamma_T'\mathbf{u}^{\mathrm{T}}|\boldsymbol{\delta}|,\\
&\text{subj. to } \sum_{j=1}^{N}\Big|\widehat{a}_{ij} + \sum_{r=1}^{M}\delta_r w_{0,r,ij}\Big| < 1, \text{ with } -1 \le \sum_{i=1}^{M}\delta_i \le 1,
\end{aligned}
\tag{2.11}
$$

where $\mathbf{u} = (|\widetilde{\delta}_1|^{-1}, \ldots, |\widetilde{\delta}_M|^{-1}|)^{\mathrm{T}}$, $|\boldsymbol{\delta}| = (|\delta_1|, \ldots, |\delta_{N^2}|)^{\mathrm{T}}$. The function $\boldsymbol{\beta}(\widehat{\boldsymbol{\xi}}, \boldsymbol{\delta})$ is defined in (2.8), with $\boldsymbol{\xi}$ replaced by the sign consistent estimator $\widehat{\boldsymbol{\xi}}$. This is a penalized profile least square type of estimator, with a more direct penalized least square formulation

$$
\begin{aligned}
\widehat{\boldsymbol{\delta}} &= \arg\min_{\boldsymbol{\delta}} \frac{1}{2T}\big\|\mathbf{B}^{\mathrm{T}}\mathbf{y} - \mathbf{B}^{\mathrm{T}}\mathbf{Z}\widehat{\boldsymbol{\xi}} - \widehat{\mathbf{g}} - (\mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0 - \mathbf{H})\boldsymbol{\delta}\big\|^2 + \gamma_T'\mathbf{u}^{\mathrm{T}}|\boldsymbol{\delta}|, \text{ with}\\
\widehat{\mathbf{g}} &= T^{-1/2}N^{-a/2}\Big(\sum_{t=1}^{T}\mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}})\boldsymbol{\gamma}\Big)(\mathbf{X}^{\mathrm{T}}\mathbf{B}^v\mathbf{B}^{v\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{B}^v\mathbf{B}^{v\mathrm{T}}(\mathbf{I}_{TN} - \widehat{\mathbf{A}}^{\otimes})\mathbf{y}^v,\\
&\text{subj. to } \sum_{j=1}^{N}\Big|\widehat{a}_{ij} + \sum_{r=1}^{M}\delta_r w_{0,r,ij}\Big| < 1, \text{ with } -1 \le \sum_{i=1}^{M}\delta_i \le 1,
\end{aligned}
\tag{2.12}
$$

The tuning parameter γ_T' is different from γ_T in general, and the choice of which is discussed in Section 4. However, they can have the same rate as stated in Assumption R7 in Section 6.

With this, the estimated spatial correlation is then

$$\widehat{\rho} = \sum_{i=1}^{M} \widehat{\delta}_i.$$

Theorem 5 presents the sign-consistency and asymptotic normality of the $\widehat{\boldsymbol{\delta}}$ under certain conditions.

Remark 1. We penalize on $\boldsymbol{\xi}$ and $\boldsymbol{\delta}$ in two different problems. The reasons we do this are twofold. Firstly, penalizing on both $\boldsymbol{\xi}$ and $\boldsymbol{\delta}$ at the same time makes the proof of theoretical results much harder. Second and more importantly, penalizing on both of them first result in two LASSO estimators, which are not as accurate as the adaptive LASSO estimators. Hence a second penalization is needed anyway on both $\boldsymbol{\xi}$ and $\boldsymbol{\delta}$. The computational complexities in finding the tuning parameters of two problems penalizing on two parameters will be more than that for finding the tuning parameter of penalizing just $\boldsymbol{\xi}$ first, then on $\boldsymbol{\xi}$ and $\boldsymbol{\delta}$ separately.

Finally, the estimator for $\boldsymbol{\mu}^*$ is

$$\widehat{\boldsymbol{\mu}} = \left(\mathbf{I}_N - \widehat{\mathbf{W}}\right)\bar{\mathbf{y}} - \bar{\mathbf{X}}\widetilde{\boldsymbol{\beta}}, \quad \text{where } \widehat{\mathbf{W}} = \widehat{\mathbf{A}} + \sum_{i=1}^{M} \widehat{\delta}_i \mathbf{W}_{0i}, \tag{2.13}$$

and $\widehat{\mathbf{A}}$ is the sparse adjustment matrix reconstructed from $\widehat{\boldsymbol{\theta}} = (\widehat{\boldsymbol{\xi}}^{\mathrm{T}}, \widehat{\boldsymbol{\delta}}^{\mathrm{T}})^{\mathrm{T}}$. The estimator for the spatial fixed effects is then

$$\left(\mathbf{I}_N - \widehat{\mathbf{W}}\right)^{-1}\widehat{\boldsymbol{\mu}} = \bar{\mathbf{y}} - \left(\mathbf{I}_N - \widehat{\mathbf{W}}\right)^{-1}\bar{\mathbf{X}}\widetilde{\boldsymbol{\beta}}.$$

3 Properties of the Estimators

We first present the notations used hereafter and introduce the measure of time dependence of all the time series variables involved. The concept of approximate sparseness of the sparse adjustment matrix $\mathbf{A}^*$ in (2.3) is then introduced in Section 3.1, together with the main assumptions in the paper. More technically involved regularity conditions are introduced in Section 6 of the paper. All theorems are presented in Section 3.4.

Denote $\{\mathbf{b}_t\} = \{\mathrm{vec}(\mathbf{B}_t)\}$ and $\mathbf{x}_t = \{\mathrm{vec}(\mathbf{X}_t)\}$ the vectorized processes for $\{\mathbf{B}_t\}$ and $\{\mathbf{X}_t\}$ respectively, both with length NK. For $t = 1, \ldots, T$, we assume that

$$\mathbf{x}_t = [f_j(\mathcal{F}_t)]_{1\le j\le NK}, \quad \mathbf{b}_t = [g_j(\mathcal{G}_t)]_{1\le j\le NK}, \quad \boldsymbol{\epsilon}_t = [h_j(\mathcal{H}_t)]_{1\le j\le NK}, \tag{3.1}$$

where the $f_j(\cdot)$'s, $g_j(\cdot)$'s and $h_j(\cdot)$'s are measurable functions defined on the real line, and $\mathcal{F}_t =$

$(\dots,\mathbf{e}_{x,t-1},\mathbf{e}_{x,t})$, $\mathcal{G}_t = (\dots,\mathbf{e}_{b,t-1},\mathbf{e}_{b,t})$, $\mathcal{H}_t = (\dots,\mathbf{e}_{\epsilon,t-1},\mathbf{e}_{\epsilon,t})$ are defined by independent and identically distributed processes $\{\mathbf{e}_{x,t}\}$ $\{\mathbf{e}_{b,t}\}$ and $\{\mathbf{e}_{\epsilon,t}\}$ respectively, with $\{\mathbf{e}_{b,t}\}$ independent of $\{\mathbf{e}_{\epsilon,t}\}$ but correlated with $\{\mathbf{e}_{x,t}\}$.

We use the functional dependence measure introduced in Wu (2005) for gauging the serial dependence of a process. For $d > 0$, define

$$\begin{aligned}
\theta^x_{t,d,j} &= \|x_{tj} - x'_{tj}\|_d = (E|x_{tj} - x'_{tj}|^d)^{1/d},\\
\theta^b_{t,d,j} &= \|b_{tj} - b'_{tj}\|_d = (E|b_{tj} - b'_{tj}|^d)^{1/d},\\
\theta^\epsilon_{t,d,\ell} &= \|\epsilon_{t\ell} - \epsilon'_{t\ell}\|_d = (E|\epsilon_{t\ell} - \epsilon'_{t\ell}|^d)^{1/d},
\end{aligned} \tag{3.2}$$

where $j = 1,\dots,NK$, $\ell = 1,\dots,N$, and $x'_{tj} = f_j(\mathcal{F}'_t)$, $\mathcal{F}'_t = (\dots,\mathbf{e}_{x,-1},\mathbf{e}'_{x,0},\mathbf{e}_{x,1},\dots,\mathbf{e}_{x,t})$, with $\mathbf{e}'_{x,0}$ independent of all other $\mathbf{e}_{x,j}$'s. Hence x'_{tj} is a coupled version of x_{tj} with $\mathbf{e}_{x,0}$ replaced by an i.i.d. copy $\mathbf{e}'_{x,0}$. Intuitively, a large $\theta^x_{t,d,j}$ means that serial correlation is strong at least for variables at most time t apart. Finally, we have similar definitions for b'_{tj} and $\epsilon'_{t\ell}$. Such a definition of "physical" or functional dependence of time series on past "inputs" is used in various papers, for example in Shao (2010) and Zhou (2010).

3.1 Main assumptions, notations and partial sign consistency

We present the main assumptions in this section. They concern the elements in the sparse adjustment matrix $\mathbf{A}^*$, the potential specifications $\mathbf{W}_{0i}$, the true spatial weight matrix $\mathbf{W}^*$ in (2.3) and the distributional behavior and serial dependence of all the time series variables involved. The identification condition of model (2.7) is also presented, with the identification of model (2.7) proved in Section 3.2. The partial sign consistency of an estimator for an approximately sparse matrix is then discussed, followed by explanations of our main assumptions. More technically involved regularity conditions including rates of convergence for various quantities are presented in Section 6.

M1. With the true spatial weight matrix $\mathbf{W}^*$ defined in (2.3), there exists a constant $\eta > 0$ such that $\|\mathbf{W}^*\|_\infty < \eta < 1$ uniformly as $N \to \infty$. The elements in $\mathbf{W}^*$ can be negative and $\mathbf{W}^*$ can be asymmetric.

M2. (Approximate sparseness for $\mathbf{A}^*$) There exists a constant $\tau > 0$ such that the elements a^*_{ij} of $\mathbf{A}^*$ are constants as $N \to \infty$ whenever they are larger than or equal to τ in magnitude. For those elements smaller than τ, we have either $a^*_{ij} = 0$ or $a^*_{ij} \to 0$ as $N \to \infty$. Moreover, the number of elements belonging to J_1 in each row of $\mathbf{A}^*$ is bounded uniformly away from infinity, where J_1 is defined in (3.3) below.

The rates in which the individual or the sum of these "small" elements satisfy will be explicitly spelt out in Assumptions R8 in Section 6.

M3. The processes $\{\mathbf{B}_t\}$, $\{\mathbf{X}_t\}$ and $\{\boldsymbol{\epsilon}_t\}$ defined in Section 2 are second-order stationary, with $\{\mathbf{X}_t\}$ and $\{\boldsymbol{\epsilon}_t\}$ having zero means. The exogenous variables $\{\mathbf{B}_t\}$ are independent of the noise $\{\boldsymbol{\epsilon}_t\}$. The tail condition $P(|Z| > v) \leq D_1 \exp(-D_2 v^q)$ is satisfied for the variables $B_{t,jk}, X_{t,jk}, \epsilon_{t,j}$ by the same constants D_1, D_2 and q.

M4. Define

$$\Theta^x_{m,a} = \sum_{t=m}^{\infty} \max_{1\leq j\leq NK} \theta^x_{t,a,j}, \;\; \Theta^b_{m,a} = \sum_{t=m}^{\infty} \max_{1\leq j\leq NK} \theta^b_{t,a,j}, \;\; \Theta^\epsilon_{m,a} = \sum_{t=m}^{\infty} \max_{1\leq j\leq NK} \theta^\epsilon_{t,a,j},$$

where $\theta^x_{t,a,j}, \theta^b_{t,a,j}$ and $\theta^\epsilon_{t,a,j}$ are defined in (3.2). The we assume that for some $w > 2$, $\Theta^x_{m,2w}, \Theta^b_{m,2w}, \Theta^\epsilon_{m,2w} \leq Cm^{-\alpha}$ with $\alpha, C > 0$ being constants that can depend on w.

M5. (Identification condition) Assume that the two sets of parameters $(\boldsymbol{\xi}^*, \boldsymbol{\delta}^*, \boldsymbol{\beta}^*)$ and $(\boldsymbol{\xi}^o, \boldsymbol{\delta}^o, \boldsymbol{\beta}^o)$ both satisfy model (2.7). Let both $\boldsymbol{\xi}^*$ and $\boldsymbol{\xi}^o$ be exactly sparse (see also Assumption M2), with the set S defined as

$$S = \{j : \xi^*_j \neq 0 \text{ or } \xi^o_j \neq 0\}.$$

Then the identification condition is that the matrix $\mathbf{Q}^{\mathrm{T}}\mathbf{Q}$ has all its eigenvalues uniformly bounded away from 0, where

$$\begin{aligned}
&\mathbf{Q} = [T^{-1/2}E(\mathbf{B}^{\mathrm{T}}\mathbf{Z}_S),\; T^{-1/2}E(\mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0),\; T^{-1/2}E(\mathbf{B}^{\mathrm{T}}\widetilde{\mathbf{X}})] \;\text{ and}\\
&\widetilde{\mathbf{X}} = (\mathbf{x}_{1,1}, \ldots, \mathbf{x}_{T,1}, \ldots, \mathbf{x}_{1,N}, \ldots, \mathbf{x}_{T,N})^{\mathrm{T}}.
\end{aligned}$$

The notation A_S means that the matrix A has columns restricted to the set S, while $\mathbf{x}^{\mathrm{T}}_{t,j}$ is the jth row of $\mathbf{X}_t$.

In addition, for approximately sparse $\boldsymbol{\xi}^*$ and $\boldsymbol{\xi}^o$, we assume that those elements which are $o(1)$ are all identified exactly to 0.

To present the concept of partial sign consistency of an estimator, which is closely related to the identification condition M5 above, we need more definitions. With respect to Assumption M2, using the notations in (2.5), define

$$\begin{aligned}
&J_0 = \{j : \xi^*_j = 0 \text{ and not corr. to the diagonal of } \mathbf{A}^*\},\\
&J_1 = \{j : |\xi^*_j| \geq \tau\},\; J_2 = \{j : 0 < |\xi^*_j| < \tau\}.
\end{aligned} \tag{3.3}$$

The set J_0 is the set for all the zeros in the sparse adjustment matrix $\mathbf{A}^*$ excluding the diagonal, which are all 0 by definition. Both J_1 and J_2 are for all the non-zeros, but those elements in J_2, according to assumption M2, are all $o(1)$ as both $T, N \to \infty$. A partially sign consistent estimator will estimate all the elements in J_0 and J_2 to be exactly 0 while estimating those in J_1 to be non-zero with the correct sign. Such a regularized estimator can accumulate smaller errors than those allowing for non-zero estimates for the small entries, since the estimation errors involved for estimating all those small entries can be much larger than setting them to zero. We show in Theorem 3 that $\widehat{\boldsymbol{\xi}}$ in (2.10) is such an estimator with probability going to 1.

We now explain briefly the assumptions M1-M4. Assumption M5 will be used in the Section 3.2 to prove the identification of model (2.7).

Assumption M1 ensures that model (2.1) has a stable reduced form

$$\mathbf{y}_t = \boldsymbol{\Pi}^*\boldsymbol{\mu}^* + \boldsymbol{\Pi}^*\mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\Pi}^*\boldsymbol{\epsilon}_t, \quad \boldsymbol{\Pi}^* = (\mathbf{I}_N - \mathbf{W}^*)^{-1}, \tag{3.4}$$

where the innovations $\boldsymbol{\Pi}^*\boldsymbol{\epsilon}_t$ have finite variances. See for example Corrado and Fingleton (2011) for a similar row sum condition in a slightly different spatial model specification. Assumption M2 is a relaxation to strict sparseness on $\mathbf{A}^*$. When $\mathbf{A}^*$ is required to be exactly sparse, it means that the specifications $\mathbf{W}_{0i}$ have to form a linear combination exactly equals to $\mathbf{W}^*$ except for $n = |J_1|$ number of elements specified in $\mathbf{A}^*$, with n not being too large. However, this is not always possible. Allowing $\mathbf{A}^*$ to have very small non-zero elements extends the scope of our estimators. See Theorems 1 and 6, and the rates of how those small elements can grow in Assumption R8. Restricting the number of large elements in each row of $\mathbf{A}^*$, together with the identification condition M5, are important in the identification of model (2.7). See Section 3.2 for more details.

The independence of $\{\mathbf{B}_t\}$ and $\{\boldsymbol{\epsilon}_t\}$ in Assumption M3 ensures that $\{\mathbf{B}_t\}$ serves a function similar to an instrument for model (2.1). The tail condition in M3 implies that all the random variables involved are with sub-exponential tails.

The assumption $\Theta^x_{m,2w} \leq Cm^{-\alpha}$ in M4 essentially means that the strongest serial dependence for the x_{tj}'s with at least m time units apart is decaying polynomially as m increases. Together with M3, they allow for the application of a Nagaev-type inequality in Lemma 1 in the supplementary material for our results to hold. Stationary Markov Chains and stationary linear processes are examples of time series that satisfy M4. See Chen et al. (2013).

3.2 Identification of the model

We show that Assumption M5 is sufficient for the identification of model (2.7). Consider two set of parameters $(\boldsymbol{\xi}^*, \boldsymbol{\delta}^*, \boldsymbol{\beta}^*)$ and $(\boldsymbol{\xi}^o, \boldsymbol{\delta}^o, \boldsymbol{\beta}^o)$ both satisfying model (2.7). Then

$$\mathbf{0} = \mathbf{B}^{\mathrm{T}}\mathbf{Z}(\boldsymbol{\xi}^* - \boldsymbol{\xi}^o) + \mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0(\boldsymbol{\delta}^* - \boldsymbol{\delta}^o) + \mathbf{B}^{\mathrm{T}}\mathbf{X}_{\boldsymbol{\beta}^* - \boldsymbol{\beta}^o}\mathrm{vec}(\mathbf{I}_N).$$

But

$$T^{-1/2}\mathbf{B}^{\mathrm{T}}\mathbf{X}_{\boldsymbol{\beta}^*-\boldsymbol{\beta}^o}\mathrm{vec}(\mathbf{I}_N) = N^{-a/2}\begin{pmatrix} T^{-1}\sum_{t=1}^T(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\mathbf{x}_{t,1}^{\mathrm{T}}(\boldsymbol{\beta}^*-\boldsymbol{\beta}^o) \\ \vdots \\ T^{-1}\sum_{t=1}^T(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\mathbf{x}_{t,N}^{\mathrm{T}}(\boldsymbol{\beta}^*-\boldsymbol{\beta}^o) \end{pmatrix}$$
$$= T^{-1/2}\mathbf{B}^{\mathrm{T}}\widetilde{\mathbf{X}}(\boldsymbol{\beta}^*-\boldsymbol{\beta}^o).$$

Hence with the definition of the set S in Assumption M5, we have

$$[T^{-1/2}\mathbf{B}^{\mathrm{T}}\mathbf{Z}_S \;\; T^{-1/2}\mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0 \;\; T^{-1/2}\mathbf{B}^{\mathrm{T}}\widetilde{\mathbf{X}}]\begin{pmatrix} \boldsymbol{\xi}_S^*-\boldsymbol{\xi}_S^o \\ \boldsymbol{\delta}^*-\boldsymbol{\delta}^o \\ \boldsymbol{\beta}^*-\boldsymbol{\beta}^o \end{pmatrix} = \mathbf{0}.$$

The above is true even for approximately sparse $\mathbf{A}^*$ and $\mathbf{A}^o$, since we have the assumption that all the $o(1)$ elements (i.e., all those elements in J_2) are identified to 0, meaning that if both ξ_j^* and ξ_j^o are in $o(1)$, then it is identified that $\xi_j^*-\xi_j^o=0$, so that the corresponding column in the matrix $\mathbf{B}^{\mathrm{T}}\mathbf{Z}$ can be removed. Hence taking expectation and multiplying $\mathbf{Q}^{\mathrm{T}}$ on both sides and then $(\mathbf{Q}^{\mathrm{T}}\mathbf{Q})^{-1}$, we have shown that $\boldsymbol{\xi}_S^*=\boldsymbol{\xi}_S^o$, $\boldsymbol{\delta}^*=\boldsymbol{\delta}^o$ and $\boldsymbol{\beta}^*=\boldsymbol{\beta}^o$.

It may seem that the assumption of being able to identify the "small" elements to 0 is strong under approximate sparsity. However, Theorem 3 does present a sign consistent estimator which estimate all "small" elements to 0 with probability going to 1. Hence such an assumption is reasonable in the face of this theoretical result for our estimator.

Note that the matrix $\mathbf{Q}$ has size $N^2\times(|S|+M+K)$. By Assumption M2 where there are only finite number of elements in each row of $\mathbf{A}^*$ and $\mathbf{A}^o$ belonging to J_1, we can see that $|S|$ is of order N. Hence assuming $\mathbf{Q}$ is of full rank is reasonable since N^2 is then much larger than $|S|+M+K$, which is also of order N. In this sense, we see that the identification of the model parameters relies mainly on the sparsity of the sparse adjustment matrix.

3.3 Allowing past $\mathbf{y}_t$ in $\mathbf{X}_t$

The assumptions above allow $\mathbf{X}_t$ to contain past values of $\mathbf{y}_t$ as long as there are exogenous $\{\mathbf{B}_t\}$. For instance, if $\mathbf{X}_t=(\mathbf{y}_{t-1},\ldots,\mathbf{y}_{t-d},\mathbf{z}_t)$ where $\mathbf{z}_t$ contains covariates other than $\{\mathbf{y}_t\}$, then

$$\mathbf{X}_t\boldsymbol{\beta}^* = \sum_{j=1}^d \beta_j^*\mathbf{y}_{t-j} + \mathbf{z}_t\boldsymbol{\beta}_2^*,$$

where $\boldsymbol{\beta}^*=(\beta_1^*,\ldots,\beta_d^*,\boldsymbol{\beta}_2^{*\mathrm{T}})^{\mathrm{T}}$. Hence model (2.1) becomes

$$\mathbf{y}_t = \boldsymbol{\mu}^* + \mathbf{W}^*\mathbf{y}_t + \sum_{j=1}^d \beta_j^*\mathbf{y}_{t-j} + \mathbf{z}_t\boldsymbol{\beta}_2^* + \boldsymbol{\epsilon}_t.$$

The reduced form model in (3.4) then becomes

$$\mathbf{y}_t = \mathbf{\Pi}^* \boldsymbol{\mu}^* + \sum_{j=1}^{d} \beta_j^* \mathbf{\Pi}^* \mathbf{y}_{t-j} + \mathbf{\Pi}^* (\mathbf{z}_t \boldsymbol{\beta}_2^* + \boldsymbol{\epsilon}_t),$$

which has a vector autoregressive part with coefficient matrices $\beta_j^* \mathbf{\Pi}^*$.

3.4 Main results

We first present the rates related to our estimators for an arbitrary sparse adjustment matrix.

Theorem 1 *Let all the assumptions in Section 3.1. Moreover, let $\alpha > 1/2 - 1/w$ in Assumption M4, and $N = o(T^{w/4-1/2} \log^{w/4}(T))$. Let $\widetilde{\mathbf{A}}$ be any estimator for the sparse adjustment $\mathbf{A}^*$ (not necessarily a LASSO estimator). Then the estimator $\widetilde{\boldsymbol{\delta}}$ in (2.9) and $\widetilde{\boldsymbol{\beta}} = \boldsymbol{\beta}(\widetilde{\boldsymbol{\theta}})$ in (2.8), with $\widetilde{\boldsymbol{\theta}} = (\widetilde{\boldsymbol{\xi}}^{\mathrm{T}}, \widetilde{\boldsymbol{\delta}}^{\mathrm{T}})^{\mathrm{T}}$, satisfy*

$$\big\|\widetilde{\boldsymbol{\delta}} - \boldsymbol{\delta}^*\big\|_1 = O_p(\lambda_T N^{-\frac{1}{2}+\frac{1}{2w}} + N^{-1}\big\|\widetilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\big\|_1) = \big\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\big\|_1.$$

From the above bound, we can see that the quality of both $\widetilde{\boldsymbol{\delta}}$ and $\widetilde{\boldsymbol{\beta}}$ are dependent on $\widetilde{\boldsymbol{\xi}}$, and hence $\widetilde{\mathbf{A}}$. If our initial specifications $\mathbf{W}_{01}, \ldots, \mathbf{W}_{0M}$ are not sufficient to form a good linear combination for estimating the spatial weight matrix, then we need more non-zero adjustments on $\mathbf{A}^*$, and estimating too many of them is going to inflate the error bound $\big\|\widetilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\big\|_1$.

Theorem 2 *Let the assumptions in Section 3.1 and in Theorem 1 hold. Then the LASSO solution $\widetilde{\boldsymbol{\xi}}$ satisfies*

$$\begin{aligned}
\big\|\widetilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\big\|_1 &= O_p(\lambda_T N^{\frac{1}{2}+\frac{1}{2w}} + n^{1/2}\big\|\widetilde{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}^*_{J_1}\big\|), \\
\big\|\widetilde{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}^*_{J_1}\big\| &= O_P(\lambda_T(n^{1/2} + N^{\frac{1}{4}+\frac{1}{4w}}) + \lambda_T N^{\frac{a}{2}+\frac{1}{2w}}(1 + n^{1/2} N^{\frac{a-1}{2}})).
\end{aligned}$$

For the LASSO estimators $\widetilde{\boldsymbol{\beta}}$ and $\widetilde{\boldsymbol{\delta}}$,

$$\big\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\big\|_1 = O_p(\lambda_T N^{-\frac{1}{2}+\frac{1}{2w}}) = \big\|\widetilde{\boldsymbol{\delta}} - \boldsymbol{\delta}^*\big\|_1.$$

It is worth noting that while $\big\|\widetilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\big\|_1$ may not necessarily converge to 0, the L_2 error on J_1, $\big\|\widetilde{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}^*_{J_1}\big\|$, is indeed going to 0 in probability. Our empirical results confirm this.

Theorem 3 *(Oracle Property for $\widehat{\boldsymbol{\xi}}$) Let the assumptions in Section 3.1 and in Theorem 1 hold. Then with large enough T, N, with probability going to 1, we have*

$$\mathrm{sign}(\widehat{\boldsymbol{\xi}}_{J_1}) = \mathrm{sign}(\boldsymbol{\xi}^*_{J_1}), \quad \widehat{\boldsymbol{\xi}}_{J_0 \cup J_2} = \mathbf{0},$$

where $J_0 = \{j : \xi_j^* = 0\}$, and J_1, J_2 are defined in (3.3). Moreover, define the predictive dependence measures

$$P_0^b(B_{t,sk}) = E(B_{t,sk}|\mathcal{G}_0) - E(B_{t,sk}|\mathcal{G}_{-1}), \;\; P_0^\epsilon(\epsilon_{tj}) = E(\epsilon_{tj}|\mathcal{H}_0) - E(\epsilon_{tj}|\mathcal{H}_{-1}),$$

where $\mathcal{G}_t$ and $\mathcal{H}_t$ are defined just below equation (3.1). Assume

$$\sum_{t\geq 0} \max_{1\leq k\leq K} \max_{1\leq s\leq N} \big\|P_0^b(B_{t,sk})\big\| < \infty, \quad \sum_{t\geq 0} \max_{1\leq j\leq N} \big\|P_0^\epsilon(\epsilon_{tj})\big\| < \infty. \tag{3.5}$$

Then for $\mathbf{M} = (\boldsymbol{\alpha}_1, \dots, \boldsymbol{\alpha}_m)^{\mathrm{T}}$ with m finite and $\|\boldsymbol{\alpha}_i\|, \|\boldsymbol{\alpha}_i\|_1 \leq c < \infty$, we have

$$T^{1/2}(\mathbf{MR\Sigma R}^{\mathrm{T}}\mathbf{M}^{\mathrm{T}})^{-1/2}\mathbf{M}(\widehat{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}^*_{J_1} + \mathbf{G}^{-1}_{J_1J_1}\gamma_T\mathbf{g}_{J_1}) \xrightarrow{\mathcal{D}} N(\mathbf{0}, \mathbf{I}_m),$$

where $\mathbf{R} = N^{-a}\mathbf{G}^{-1}_{J_1J_1}E(T^{-1}\mathbf{Z}^{\mathrm{T}}_{J_1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma))$, $\boldsymbol{\Sigma} = \sum_\tau E(\boldsymbol{\epsilon}_t\boldsymbol{\epsilon}^{\mathrm{T}}_{t+\tau}) \otimes E((\mathbf{B}_t - \boldsymbol{\mu}_b)\boldsymbol{\gamma}\boldsymbol{\gamma}^{\mathrm{T}}(\mathbf{B}_{t+\tau} - \boldsymbol{\mu}_b)^{\mathrm{T}})$. The vector $\mathbf{g}_{J_1}$ is a vector of 1 or -1, depending on if the corresponding element in $\boldsymbol{\xi}^*_{J_1}$ is positive or negative.

The predictive dependence measure is introduced in Definition 2 of Wu (2011) for quantifying the degree of dependence of outputs on inputs in physical systems similar to our variables defined in (3.1). To make inference on the non-zero sparse adjustments on $\mathbf{A}^*$, we can use the sign consistency of $\widehat{\boldsymbol{\xi}}$ and the asymptotic normality of $\widetilde{\boldsymbol{\xi}}_{J_1}$. We can estimate the matrices $\mathbf{R}$ and $\boldsymbol{\Sigma}$ by the corresponding sample autocovariance estimators. Since $\{\boldsymbol{\epsilon}_t\}$ is unobserved, we replace this by $\{\widehat{\boldsymbol{\epsilon}}_t\}$ obtained as the residuals of model (2.1) with $\boldsymbol{\mu}^*$, $\mathbf{A}^*$, $\boldsymbol{\delta}^*$ and $\boldsymbol{\beta}^*$ replaced by $\widehat{\boldsymbol{\mu}}$, $\widehat{\mathbf{A}}$, $\widehat{\boldsymbol{\delta}}$ and $\widehat{\boldsymbol{\beta}}$ respectively. Note that $\mathbf{R}$ is in fact independent of the unknown index a since $\mathbf{G}$ has the term N^{-a} as well, canceling out the N^{-a} in the definition of $\mathbf{R}$. Since the term $\boldsymbol{\xi}^*_{J_1}$ is a vector of non-zero constants while on $\mathcal{M}$, $\big\|\mathbf{G}^{-1}_{J_1J_1}\gamma_T\mathbf{g}_{J_1}\big\|_{\max} = O(\lambda_T) = o(1)$ (see the proof of the theorem in the supplementary materials for more details), we can ignore the term $\mathbf{G}^{-1}_{J_1J_1}\gamma_T\mathbf{g}_{J_1}$ in using the asymptotic normality result in construction of confidence intervals.

In practice, we can find a reasonable cut-off on τ in calculating $\boldsymbol{\Sigma}$. See Section 4 for more practical details on this. The rate of convergence of each $\widehat{\boldsymbol{\xi}}_j$ for $j \in J_1$ can be deduced to be $T^{-1/2}N^{-(a-b)/2}$ from the asymptotic normality by using Assumption R6 in Section 6 to deduce that $N^{-b}\boldsymbol{\Sigma}$ has uniformly bounded eigenvalues from 0 and infinity. Simulation results can be found in Section 5.

With this, we present the asymptotic normality for $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}(\widehat{\boldsymbol{\xi}}, \widetilde{\boldsymbol{\delta}})$ defined in (2.8), and the oracle property of $\widehat{\boldsymbol{\delta}}$ defined in (2.11) in the following two theorems.

Theorem 4 *Let the assumptions in Section 3.1 and in Theorem 1 hold. Assume that the predictive dependence measures $P_0^b(B_{tk})$ and $P_0^\epsilon(\epsilon_{tj})$ are as defined in Theorem 3 with the same assumptions (3.5) applied. Then for $\boldsymbol{\alpha}_j \in \mathbb{R}^K$, $j = 1, \dots, m$ with $m \leq K$ such that $\|\boldsymbol{\alpha}_j\| = 1$ and $\mathbf{M} =$*

$(\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_m)^{\mathrm{T}}$, *we have*

$$T^{1/2}\mathbf{S}_0^{-1/2}\mathbf{M}^{\mathrm{T}}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*) \xrightarrow{\mathcal{D}} N(\mathbf{0}, \mathbf{I}_m),$$

where, defining $\mathbf{R}_0 = (E(\mathbf{X}_t^{\mathrm{T}}\mathbf{B}_t)E(\mathbf{B}_t^{\mathrm{T}}\mathbf{X}_t))^{-1}E(\mathbf{X}_t^{\mathrm{T}}\mathbf{B}_t)$,

$$\mathbf{S}_0 = \sum_{\tau} \mathbf{M}\mathbf{R}_0 E(\mathbf{B}_t^{\mathrm{T}}\boldsymbol{\epsilon}_t\boldsymbol{\epsilon}_{t+\tau}^{\mathrm{T}}\mathbf{B}_{t+\tau})\mathbf{R}_0^{\mathrm{T}}\mathbf{M}^{\mathrm{T}}.$$

Theorem 5 *(Oracle property for $\widehat{\boldsymbol{\delta}}$) Let the assumptions in Section 3.1 and in Theorem 1 hold. We then have, as $T, N \to \infty$, with probability approaching 1,*

$$\mathrm{sign}(\widehat{\boldsymbol{\delta}}_H) = \mathrm{sign}(\boldsymbol{\delta}_H^*), \quad \widehat{\boldsymbol{\delta}}_{H^c} = \mathbf{0}, \quad \textit{where } H = \{j : \delta_j^* \neq 0\}.$$

Moreover, with the predictive dependence measures $P_0^b(B_{tk})$ and $P_0^{\epsilon}(\epsilon_{tj})$ as defined in Theorem 3 with the same assumptions (3.5) applied, we have

$$T^{1/2}(\mathbf{R}_1\mathbf{S}_{\boldsymbol{\gamma}}\mathbf{S}_0\mathbf{S}_{\boldsymbol{\gamma}}^{\mathrm{T}}\mathbf{R}_1^{\mathrm{T}})^{-1/2}(\widehat{\boldsymbol{\delta}}_H - \boldsymbol{\delta}_H^*) \xrightarrow{\mathcal{D}} N(\mathbf{0}, \mathbf{I}_{|H|}),$$

where $\mathbf{R}_1 = [(\mathbf{H}_{10} - \mathbf{H}_{20})_H^{\mathrm{T}}(\mathbf{H}_{10} - \mathbf{H}_{20})_H]^{-1}(\mathbf{H}_{10} - \mathbf{H}_{20})_H^{\mathrm{T}}$, *with* $\mathbf{H}_{10}, \mathbf{H}_{20}$ *and* $\mathbf{S}_{\boldsymbol{\gamma}}$ *defined as*

$$\begin{aligned}
\mathbf{H}_{10} &= \Big(\mathbf{I}_N \otimes (\mathbf{I}_N \otimes \boldsymbol{\gamma}^{\mathrm{T}})E(vec(\mathbf{B}_t^{\mathrm{T}})vec(\mathbf{X}_t^{\mathrm{T}})^{\mathrm{T}})(\mathbf{I}_N \otimes \boldsymbol{\beta}^*)\boldsymbol{\Pi}^{*\mathrm{T}}\Big)\mathbf{V}_0,\\
\mathbf{H}_{20} &= E(\mathbf{X}_t \otimes \mathbf{B}_t\boldsymbol{\gamma})\Big(E(\mathbf{X}_t^{\mathrm{T}}\mathbf{B}_t)E(\mathbf{B}_t^{\mathrm{T}}\mathbf{X}_t)\Big)^{-1}E(\mathbf{X}_t^{\mathrm{T}}\mathbf{B}_t)\Big(\mathbf{V}_{\mathbf{W}_{01}^{\mathrm{T}}}^{\mathrm{T}} \cdots \mathbf{V}_{\mathbf{W}_{0M}^{\mathrm{T}}}^{\mathrm{T}}\Big)(\mathbf{I}_M \otimes \mathbf{U}_0\mathbf{V}_{\boldsymbol{\Pi}^*}\boldsymbol{\beta}^*),\\
\mathbf{S}_{\boldsymbol{\gamma}} &= \Big(cov(\mathbf{x}_{t,1}, \mathbf{b}_{t,1})\boldsymbol{\gamma}, \ldots, cov(\mathbf{x}_{t,1}, \mathbf{b}_{t,N})\boldsymbol{\gamma}, \ldots, cov(\mathbf{x}_{t,N}, \mathbf{b}_{t,1})\boldsymbol{\gamma}, \ldots, cov(\mathbf{x}_{t,N}, \mathbf{b}_{t,N})\boldsymbol{\gamma}\Big)^{\mathrm{T}}.
\end{aligned}$$

The definition of $\mathbf{S}_0$ is as in Theorem 4, with $\mathbf{M} = \mathbf{I}_K$. And we also assumed that the smallest eigenvalue of $\mathbf{R}_1\mathbf{S}_{\boldsymbol{\gamma}}\mathbf{S}_{\boldsymbol{\gamma}}^{\mathrm{T}}\mathbf{R}_1^{\mathrm{T}}$ is of constant order. In particular, we then have

$$T^{1/2}s_1^{-1/2}(\widehat{\rho} - \rho^*) = T^{1/2}s_1^{-1/2}\mathbf{1}_{|H|}^{\mathrm{T}}(\widehat{\boldsymbol{\delta}}_H - \boldsymbol{\delta}_H^*) \xrightarrow{\mathcal{D}} N(0, 1),$$

where $s_1 = \mathbf{1}_{|H|}^{\mathrm{T}}\mathbf{R}_1\mathbf{S}_{\boldsymbol{\gamma}}\mathbf{S}_0\mathbf{S}_{\boldsymbol{\gamma}}^{\mathrm{T}}\mathbf{R}_1^{\mathrm{T}}\mathbf{1}_{|H|}$.

The above theorem is important in determining which potential specifications of spatial weight matrices are important and which are not, and how important each specification is. Similar to $\widehat{\boldsymbol{\xi}}_{J_1}$, hypothesis testing and confidence regions can be constructed for the elements in $\widehat{\boldsymbol{\beta}}$ and $\widehat{\boldsymbol{\delta}}$ through estimating various autocovariance matrices from the data using their respective asymptotic normality.

It may seem that assuming the smallest eigenvalue of $\mathbf{R}_1\mathbf{S}_{\boldsymbol{\gamma}}\mathbf{S}_{\boldsymbol{\gamma}}^{\mathrm{T}}\mathbf{R}_1^{\mathrm{T}}$ having a constant order is a strong one. However, in the proof of Theorem 5 in the supplementary material, we have shown that the largest eigenvalue of $\mathbf{R}_1\mathbf{S}_{\boldsymbol{\gamma}}\mathbf{S}_{\boldsymbol{\gamma}}^{\mathrm{T}}\mathbf{R}_1^{\mathrm{T}}$ has a constant order. Considering the matrix is $K \times K$ which has a constant size, this assumption is not a particularly strong one.

Theorem 6 *Let the assumptions in Section 3.1 and in Theorem 1 hold. Then*

$$\|\widehat{\mathbf{W}} - \mathbf{W}^*\| = O_p(\lambda_T) = \|\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}^*\|_{\max}.$$

The rate for the spatial fixed effect, $\|\widehat{\mathbf{\Pi}}\widehat{\boldsymbol{\mu}} - \mathbf{\Pi}^*\boldsymbol{\mu}^*\|_{\max}$, *is the same.*

The above gives a rate of convergence for the spectral norm of the error for the estimated spatial weight matrix, which can be of independent interest.

Remark 2. Assumption R8 in Section 6 specifies how large N can be compared with T. For instance, if $\mathbf{A}^* = \mathbf{0}$ (which is likely to be the case in our real data application when the specified spatial weight matrices are of good quality; see Section 5.2), then $n = 0$, and all the rates in R8 reduce to only $\lambda_T N^{1-a}, T^{-1/2}N^{\frac{a-b}{2}}\log(T \vee N) = o(1)$ and $T^{-1/2}N^{b/2+1/(2w)} = o(1)$. Suppose $a = b = 3/4$, then the above further reduce to just $\lambda_T N^{1/4} = o(1)$ and $T^{-1/2}N^{3/8+1/(2w)} = o(1)$, which is satisfied when $N = o(T^{4/3})$ for instance. So N can indeed be growing faster than T under suitable conditions.

4 Practical Implementation

In this section we provide details of the block-coordinate descent (BCD) algorithm used to conduct the minimization of (2.8)-(2.12). As in Lam and Souza (2015b), the optimization problem with respect to $(\boldsymbol{\xi}, \boldsymbol{\beta}, \boldsymbol{\delta})$ is not convex and, as a consequence, significant computational challenges arise.

We make use of the fact that fixing any two of the three variables, the problem is convex in the remaining variable. For example, given $(\boldsymbol{\xi}, \boldsymbol{\beta})$, the optimization problem is convex in $\boldsymbol{\delta}$. While it is difficult to establish global convergence of the BCD algorithm without convexity, each iteration delivers an improvement of the objective functions since given one parameter, the objective functions are convex in the others. The BCD algorithm is closely related to the Iterative Coordinate Descent of Fan and Lv (2011), and is also discussed in Friedman et al. (2010) and Dicker et al. (2013).

The Block Coordinate Descent Algorithm (LASSO stage)

0. Set an initial value $\boldsymbol{\xi}^{(0)}$, for example $\boldsymbol{\xi}^{(0)} = 0$.

1. At iteration r, update $\boldsymbol{\beta}^{(r)}$ according to the closed-form expression

$$\boldsymbol{\beta}^{(r)} = (\mathbf{X}^{\mathrm{T}}\mathbf{B}^{v}\mathbf{B}^{v\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{B}^{v}\mathbf{B}^{v\mathrm{T}}\Big(\mathbf{I}_{TN} - (\mathbf{I}\otimes\mathbf{A}^{(r)}) - \sum_{i=1}^{M}\delta_i^{(r-1)}\mathbf{W}_{0i}^{\otimes}\Big)\mathbf{y}^{v}.$$

2. Update $\boldsymbol{\delta}^{(r)}$ as a function of $\boldsymbol{\beta}^{(r)}$ and $\boldsymbol{\xi}^{(r)}$, with

$$\begin{aligned}\boldsymbol{\delta}^{(r)} &= \Big[(\mathbf{H}-\mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0)^{\mathrm{T}}(\mathbf{H}-\mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0)\Big]^{-1}(\mathbf{H}-\mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0)^{\mathrm{T}} \\ &\quad \cdot(\mathbf{B}^{\mathrm{T}}\mathbf{Z}\boldsymbol{\xi}^{(r)}-\mathbf{B}^{\mathrm{T}}\mathbf{y}+\mathbf{K}(\mathbf{I}_{TN}-(\mathbf{I}-\mathbf{A}^{(r)}))\mathbf{y}^{v}).\end{aligned}$$

3. Using the Least Angle Regression (LARS), solve sequentially for each row of $\mathbf{A}$, denoted η_j, fixing all the remaining $(1,\dots,j-1,j+1,\dots,n)$ rows, for $j=1,\dots,N$. That is, solve

$$\eta_j^{(r)} = \arg\min_{\eta_j}\frac{1}{2T}\big\|\mathbf{B}^{\mathrm{T}}\mathbf{y}-\mathbf{B}^{\mathrm{T}}\mathbf{Z}\eta-\mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0\boldsymbol{\delta}^{(r-1)}-\mathbf{B}^{\mathrm{T}}\mathbf{X}_{\boldsymbol{\beta}^{(r-1)}}\mathrm{vec}(\mathbf{I}_N)\big\|^2+\gamma_T\big\|\eta\big\|_1,$$

   where $\eta=(\eta_1^{(r-1)\mathrm{T}},\dots,\eta_{j-1}^{(r-1)\mathrm{T}},\eta_j,\eta_{j+1}^{(r-1)\mathrm{T}},\dots,\eta_n^{(r-1)\mathrm{T}})^{\mathrm{T}}$, and subject to the constraints as stated in (2.8). We obtain $\boldsymbol{\xi}^{(r)}=(\eta_1^{(r)},\dots,\eta_n^{(r)})$ and, similarly, $\mathrm{vec}(\mathbf{A}^{(r)})=\boldsymbol{\xi}^{(r)}$.

4. Repeat steps 1-3, until $\big\|\xi^{(r)}-\xi^{(r-1)}\big\|$ is smaller than some predetermined number. The LASSO solution is $\widetilde{\boldsymbol{\xi}}=\boldsymbol{\xi}^{(r)}$, $\widetilde{\boldsymbol{\beta}}=\boldsymbol{\beta}^{(r)}$ and $\widetilde{\boldsymbol{\delta}}=\boldsymbol{\delta}^{(r)}$.

With the LASSO solution, we are able to compute the final adaptive LASSO stage, which is obtained by setting the initial values as $\widetilde{\boldsymbol{\xi}}$, $\widetilde{\boldsymbol{\beta}}$ and $\widetilde{\boldsymbol{\delta}}$, and repeating the steps above with the following modifications:

<u>The Block Coordinate Descent Algorithm</u> (Adaptive LASSO stage)

2′. Introduce penalization for δ:

$$\boldsymbol{\delta}^{(r)} = \arg\min_{\boldsymbol{\delta}}\frac{1}{2T}\big\|\mathbf{B}^{\mathrm{T}}\mathbf{y}-\mathbf{B}^{\mathrm{T}}\mathbf{Z}\boldsymbol{\xi}^{(r-1)}-\mathbf{g}^{(r-1)}-(\mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0-\mathbf{H})\boldsymbol{\delta}\big\|^2+\gamma_T'\mathbf{u}^{\mathrm{T}}|\boldsymbol{\delta}|,$$

   where $\mathbf{u}=(|\widetilde{\delta}_1|^{-1},\dots,|\widetilde{\delta}_M|^{-1})^{\mathrm{T}}$, $\mathbf{g}^{(r-1)}$ is defined similar to $\widehat{\mathbf{g}}$ in (2.12) with $\widehat{\mathbf{A}}$ replaced by $\mathbf{A}^{(r-1)}$, and the above is subjected to the constraints as stated in (2.12).

3′. The adaptive LASSO objective function observes penalty $\gamma_T\mathbf{v}^{\mathrm{T}}|\eta|$, that is,

$$\eta_j^{(r)} = \arg\min_{\eta_j}\frac{1}{2T}\big\|\mathbf{B}^{\mathrm{T}}\mathbf{y}-\mathbf{B}^{\mathrm{T}}\mathbf{Z}\eta-\mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0\boldsymbol{\delta}^{(r-1)}-\mathbf{B}^{\mathrm{T}}\mathbf{X}_{\boldsymbol{\beta}^{(r-1)}}\mathrm{vec}(\mathbf{I}_N)\big\|^2+\gamma_T\mathbf{v}^{\mathrm{T}}|\eta|$$

   where $\mathbf{v}=(|\widetilde{\xi}_1|^{-1},\dots,|\widetilde{\xi}_{N^2}|^{-1})^{\mathrm{T}}$, and remaining definitions as in step 3. The above is also subjected to the constraints as stated in (2.10).

For the choice of γ_T and γ_T' used in the penalization of $\boldsymbol{\xi}$ and $\boldsymbol{\delta}$ respectively, we minimize the following BIC criterion with respect to both of them,

$$\begin{aligned}\mathrm{BIC}(\gamma_{\xi,T},\gamma_{\delta,T}) &= \log\left(T^{-2}\Big\|\mathbf{B}^{\mathrm{T}}\mathbf{y}-\mathbf{B}^{\mathrm{T}}\mathbf{Z}\widehat{\boldsymbol{\xi}}-\mathbf{B}^{\mathrm{T}}\mathbf{Z}\mathbf{V}_0\widehat{\boldsymbol{\delta}}-\mathbf{B}^{\mathrm{T}}\mathbf{X}_{\widehat{\boldsymbol{\beta}}}\mathrm{vec}(\mathbf{I}_N)\Big\|^2\right) \\ &\quad +|\widehat{S}|\frac{\log(T)}{T}\log(\log(2N-2)) \qquad (4.6)\end{aligned}$$

where $\widehat{\boldsymbol{\xi}}$ and $\widehat{\boldsymbol{\delta}}$ are the Adaptive Lasso solutions with tuning parameters $\gamma_{\xi,T}$ and $\gamma_{\delta,T}$ respectively, and $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}(\widehat{\boldsymbol{\xi}}, \widehat{\boldsymbol{\delta}})$. The set $\widehat{S}$ is the indices for the non-zero values of $\widehat{\boldsymbol{\xi}}$. The BIC criterion (4.6) is in fact inspired by Wang et al. (2009), and is very similar to the one used in Lam and Souza (2015b). Although the value of a is unknown in the definition of $\mathbf{B}$, the optimal values γ_T and γ_T' are in fact independent of a because of the logarithmic operation in the first term of (4.6). In practice we set $a = 1$.

4.1 Two-stage least squares extension

We have set $\boldsymbol{\gamma} = L^{-1}\mathbf{1}_L$ in (2.6) as fixed. In fact this can be estimated to provide maximal correlation with the endogeneous variable $\mathbf{y}_t$ through two-stage least squares. In doing so, we consider the model

$$\mathbf{y}_t = \boldsymbol{\alpha} + \mathbf{B}_t\boldsymbol{\gamma} + \mathbf{v}_t,$$

where $\boldsymbol{\alpha}$ is an $N \times 1$ vector of unknown coefficients, and $\boldsymbol{\gamma}$ is the $K \times 1$ vector of coefficients we want to estimate. To get $\widehat{\boldsymbol{\gamma}}$, we can consider the problem

$$\min_{\boldsymbol{\alpha},\boldsymbol{\gamma}} \sum_{t=1}^{T} \big\|\mathbf{y}_t - \boldsymbol{\alpha} - \mathbf{B}_t\boldsymbol{\gamma}\big\|^2,$$

where we have the solution

$$\widehat{\boldsymbol{\gamma}} = \Big(\sum_{t=1}^{T}(\mathbf{B}_t - \bar{\mathbf{B}})^{\mathrm{T}}(\mathbf{B}_t - \bar{\mathbf{B}})\Big)^{-1}\sum_{t=1}^{T}(\mathbf{B}_t - \bar{\mathbf{B}})^{\mathrm{T}}(\mathbf{y}_t - \bar{\mathbf{y}}).$$

Implementing this does not really change our proof much since we can easily show that $\big\|\widehat{\boldsymbol{\gamma}}\big\|_1 = O_P(1)$, which substitute $\big\|\boldsymbol{\gamma}\big\|_1 = 1$ in all of our proofs.

We can consider more general extension for the two stage least squares, for instance, we can consider

$$\mathbf{y}_t = \boldsymbol{\alpha} + \sum_{i=1}^{N}\mathbf{A}_i\mathbf{B}_{t,i} + \mathbf{v}_t,$$

where $\mathbf{A}_i$ is an $N \times N$ coefficient matrix and $\mathbf{B}_{t,i}$ is the ith column of $\mathbf{B}_t$. A simple extension can take $\mathbf{A}_i$ to be diagonal. We leave it for future research.

5 Numerical Examples

We conduct detailed simulation exercises in Section 5.1 to demostrate the finite-sample performance of our estimators. In Section 5.2, we use our methodology to analyze the determinants of cross-country Gross Domestic Product (GDP) spillovers. From several expert spatial neighboring matrices, we find that physical distance and monetary union explain better the behavior of the series.

5.1 Simulation

Two sets of simulation exercises are conducted, one with penalization of δ, and one without. We first draw the elements of the expert neighboring matrices $\mathbf{W}_{0i}$ from an uniform distribution. If the row-sums exceed one, we divide by the L_1 norm of the row. We use five expert neighboring matrices, with $\boldsymbol{\delta}^* = (0,0,0,0,0)^{\mathrm{T}} = \mathbf{0}$, $\boldsymbol{\delta}^* = (0.2,0,0,0,0)^{\mathrm{T}} = 0.2\mathbf{e}_1$ or $\boldsymbol{\delta}^* = (0.3,0.3,0,0,0)^{\mathrm{T}} = 0.3(\mathbf{e}_1+\mathbf{e}_2)$. The first case corresponds to the scenario where no expert matrices are relevant to the problem, and the method reduces to estimating the whole spatial weight matrix; see Lam and Souza (2015b). We then add a sparse adjustment matrix $\mathbf{A}^*$. This matrix is generated by randomly selecting a row and setting one of its elements to -0.3 and another element to 0.3, ensuring that the true neighboring matrix $\mathbf{A}^* + \sum_{i=1}^{5} \mathbf{W}_{0i}$ conforms with row-sum normalization. The sparse adjustment matrix is 95% sparse. To ensure that our results are not influenced by the choice of a single true model, at each iteration the $\mathbf{W}_{0i}$ is resampled.

At each period, disturbances $\boldsymbol{\epsilon}_t$ are jointly normally distributed, with variance matrix with 1 in the diagonal. The off-diagonal elements are randomly chosen to be 0.25, introducing spatial dependence also in the disturbance term. The same procedure applies for the covariates $\mathbf{X}_t$. All simulations introduce individual fixed-effects. Finally, response data $\mathbf{y}_t$ are generated according to model (2.4).

We use several criteria to evaluate the performance of the estimator in tables 1 and 2. Specificity is defined as the proportion of true zeros estimated as zeros, while sensitivity is the proportion of non-zeros estimated as non-zeros. These measures apply to $\mathbf{A}^*$ and $\boldsymbol{\delta}^*$. Furthermore, we present the L_1 norm $\|\widetilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1$ for the LASSO estimator $\widetilde{\boldsymbol{\xi}}$, and also $\|\widehat{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1$ for the adaptive LASSO estimator $\widehat{\boldsymbol{\xi}}$. We present the same measure for $\boldsymbol{\beta}$, $\boldsymbol{\delta}$ and the fixed effects $\boldsymbol{\mu}$ only at the adaptive LASSO step.

In all simulations, we consider $N = 25$, 50 and 75 and $T = 50$, 100 and 200. Tuning parameters are chosen by the BIC criterion in (4.6). Each combination has at least 1,500 replications. Standard errors across replications are shown in parenthesis.

Table 1 shows very good performance of the estimator without penalization of δ. Note that, since no estimates of $\boldsymbol{\delta}^*$ are exactly zero, specificity is mechanically 0% and sensitivity is 100%. $\mathbf{A}^*$ specificities and sensitivities are very high, consistently above 95% except for low sample size. The biases are small and decreasing with T. The comparison between the LASSO and Adaptive LASSO steps in the estimation of $\mathbf{A}^*$ highlights the importance of the latter.

Table 2 introduces penalization of δ. Note that the evaluation criteria are not significantly affected, except for the L_1 norm $\|\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}^*\|_1$ which are significantly smaller than the corresponding values in Table 1, showing that penalization on $\boldsymbol{\delta}$ is important in getting the weights more accurate. Moreover, specificity and sensitivity for δ^* are high except for low sample sizes. These exercises highlight the good performance of the estimator in most scenario.

Next, we evaluate the empirical performance of the variance expressions in Theorems 3 and 4. From Theorem 4, we can approximate

$$T^{1/2}\widehat{\mathbf{S}}_0^{-1/2}\mathbf{M}^{\mathrm{T}}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*) \quad \approx \quad N(\mathbf{0}, \mathbf{I}_m) \tag{5.1}$$

and ignoring the term $\mathbf{G}^{-1}_{J_1 J_1}\gamma_T \mathbf{g}_{J_1}$ since $\gamma_T \to 0$, from Theorem 3, we have

$$T^{1/2}\left(\mathbf{M}\widehat{\mathbf{R}}\widehat{\boldsymbol{\Sigma}}\widehat{\mathbf{R}}^{\mathrm{T}}\mathbf{M}^{\mathrm{T}}\right)^{-1/2}\mathbf{M}(\widehat{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}^*_{J_1}) \quad \approx \quad N(\mathbf{0}, \mathbf{I}_m) \tag{5.2}$$

where the matrices $\widehat{\mathbf{S}}_0$, $\widehat{\mathbf{R}}$ and $\widehat{\boldsymbol{\Sigma}}$ are the sample counterparts of $\mathbf{S}_0$, $\mathbf{R}$ and $\boldsymbol{\Sigma}$. Since $\mathbf{S}_0$ depends on $E(\mathbf{B}_t^{\mathrm{T}}\boldsymbol{\epsilon}_t\boldsymbol{\epsilon}_{t+\tau}^{\mathrm{T}}\mathbf{B}_{t+\tau})$ for all possible τ, in empirical applications we truncate the sum to $\tau = -10, \ldots, 10$. We replace $\boldsymbol{\epsilon}_t$ by the estimated residuals $\widehat{\boldsymbol{\epsilon}}_t$ in calculating the expectation, after fitting the model with adaptive LASSO estimates. We apply a similar treatment to $\boldsymbol{\Sigma}$ as well. We found that this is more than sufficient to capture the time dependence of the data. Therefore, simulations are largely invariant to variations from this choice.

We report in Table 3 the variances of the left-hand side of equations (5.1) and (5.2) which are asymptotically normally distributed with variance 1. For small T, the variances of $\widehat{\boldsymbol{\beta}}$ and $\widehat{\xi}_{J_1}$ are larger than that implied by the expressions in Theorems 3 and 4. It means that in practice we tend to underestimate the variance of $\widehat{\boldsymbol{\beta}}$ and $\widehat{\boldsymbol{\xi}}_{J_1}$ when T is smaller relative to N. However, as T grows and N remains fixed, the variance quickly converges to 1, as implied by theory.

5.2 Application

A long-standing question in applied macroeconomics is: how shocks in a nation's Gross Domestic Product propagate to neighboring countries? Yet, what are the channels through which the GDP spillovers occur? Leading candidates are imports, exports, cross-country investments, physical distance, language commonality and belonging to a common monetary union. Given those several possibilities, how to best define a neighbor?

We explore this issue by introducing a several expert neighboring matrices, and allowing the estimator to select their combination that best describe the data.

We use GDP database of the International Monetary Fund's (IMF) World Economic Outlook report from 1980 to 2015 for the 20 largest world economies in 2015.[1] Those economies combined represent approximately 76% of global GDP at present values.[2] We first-difference data to obtain

[1]More specifically, Gross Domestic Product at constant prices

[2]The analyzed countries, sorted in descending order of GDP, are the United States, China, Japan, Germany, United Kingdom, France, Brazil, Italy, India, Canada, Australia, South Korea, Spain, Mexico, Indonesia, Netherlands, Turkey, Saudi Arabia and Switzerland. Russia was excluded due to GDP data unavailability for the entire period.

Table 1: Not penalizing $\boldsymbol{\delta}$

	$\boldsymbol{\delta}^* = \mathbf{0}$			$\boldsymbol{\delta}^* = 0.2\mathbf{e}_1$			$\boldsymbol{\delta}^* = 0.3(\mathbf{e}_1 + \mathbf{e}_2)$		
	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$
	$N = 25$								
$\mathbf{A}^*$ Specificity	95.94% (0.958)	96.59% (0.803)	97.11% (0.772)	95.79% (1.030)	96.62% (0.721)	96.99% (0.956)	95.51% (1.135)	96.19% (0.998)	96.68% (1.035)
$\mathbf{A}^*$ Sensitivity	84.01% (8.042)	98.30% (2.578)	100.00% (0.000)	84.88% (8.028)	98.29% (2.556)	99.95% (0.553)	82.39% (8.035)	98.10% (2.528)	99.96% (0.347)
$\Vert\widetilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\Vert_1$	19.118 (1.330)	17.652 (1.026)	14.444 (0.937)	22.092 (1.233)	20.354 (1.093)	17.332 (1.117)	28.268 (1.482)	26.336 (1.441)	23.675 (1.414)
$\Vert\widehat{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\Vert_1$	5.856 (0.873)	3.461 (0.519)	2.137 (0.380)	10.313 (0.853)	8.086 (0.497)	6.823 (0.413)	19.677 (0.917)	17.425 (0.585)	16.226 (0.448)
$\mathbf{A}^*$ Sparsity	0.920 (0.010)	0.919 (0.008)	0.924 (0.008)	0.919 (0.011)	0.920 (0.007)	0.923 (0.009)	0.917 (0.011)	0.915 (0.009)	0.920 (0.010)
$\Vert\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\Vert_1$	0.043 (0.021)	0.027 (0.015)	0.017 (0.010)	0.041 (0.020)	0.028 (0.014)	0.019 (0.010)	0.047 (0.026)	0.031 (0.017)	0.019 (0.011)
$\boldsymbol{\delta}^*$ Specificity	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)
$\boldsymbol{\delta}^*$ Sensitivity	−% (−)	−% (−)	−% (−)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)
$\Vert\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}^*\Vert_1$	0.064 (0.021)	0.052 (0.018)	0.049 (0.017)	0.065 (0.022)	0.051 (0.019)	0.054 (0.020)	0.067 (0.024)	0.057 (0.020)	0.056 (0.021)
$\Vert\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}^*\Vert_{\max}$	0.194 (0.039)	0.126 (0.024)	0.088 (0.018)	0.196 (0.038)	0.127 (0.026)	0.090 (0.017)	0.204 (0.039)	0.129 (0.026)	0.096 (0.021)
	$N = 50$								
$\mathbf{A}^*$ Specificity	95.15% (0.490)	96.33% (0.420)	97.20% (0.401)	95.07% (0.430)	96.28% (0.464)	97.13% (0.427)	94.96% (0.554)	96.22% (0.452)	97.10% (0.427)
$\mathbf{A}^*$ Sensitivity	86.11% (3.880)	98.93% (1.034)	100.00% (0.061)	86.72% (3.620)	98.90% (1.043)	100.00% (0.062)	86.17% (4.348)	98.97% (1.037)	99.99% (0.089)
$\Vert\widetilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\Vert_1$	66.397 (2.653)	60.583 (2.258)	53.821 (1.855)	72.799 (2.245)	66.932 (2.117)	59.609 (2.094)	85.317 (2.937)	78.071 (2.344)	70.989 (2.280)
$\Vert\widehat{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\Vert_1$	24.482 (2.057)	13.318 (1.143)	7.857 (0.684)	33.220 (1.878)	22.568 (1.248)	17.307 (0.765)	51.547 (2.247)	40.732 (1.189)	35.936 (0.829)
$\mathbf{A}^*$ Sparsity	0.912 (0.005)	0.917 (0.004)	0.925 (0.004)	0.911 (0.004)	0.917 (0.004)	0.924 (0.004)	0.910 (0.005)	0.916 (0.004)	0.924 (0.004)
$\Vert\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\Vert_1$	0.045 (0.022)	0.027 (0.014)	0.017 (0.008)	0.042 (0.020)	0.029 (0.015)	0.017 (0.009)	0.048 (0.024)	0.031 (0.013)	0.020 (0.009)
$\boldsymbol{\delta}^*$ Specificity	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)
$\boldsymbol{\delta}^*$ Sensitivity	−% (−)	−% (−)	−% (−)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)
$\Vert\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}^*\Vert_1$	0.074 (0.024)	0.064 (0.022)	0.061 (0.020)	0.072 (0.026)	0.065 (0.023)	0.060 (0.021)	0.075 (0.028)	0.072 (0.024)	0.064 (0.023)
$\Vert\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}^*\Vert_{\max}$	0.252 (0.042)	0.156 (0.026)	0.101 (0.015)	0.253 (0.035)	0.159 (0.026)	0.101 (0.015)	0.253 (0.043)	0.161 (0.027)	0.108 (0.015)
	$N = 75$								
$\mathbf{A}^*$ Specificity	95.10% (0.337)	96.69% (0.293)	97.68% (0.255)	95.17% (0.351)	96.67% (0.282)	97.63% (0.286)	94.97% (0.330)	96.56% (0.292)	97.59% (0.323)
$\mathbf{A}^*$ Sensitivity	84.25% (2.813)	98.72% (0.966)	100.00% (0.000)	84.88% (3.044)	98.74% (0.821)	100.00% (0.000)	84.91% (2.640)	98.90% (0.704)	99.99% (0.047)
$\Vert\widetilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\Vert_1$	132.259 (4.405)	117.592 (3.169)	105.960 (2.512)	142.038 (4.270)	127.249 (3.202)	116.026 (2.920)	162.667 (3.801)	147.225 (3.582)	134.257 (2.992)
$\Vert\widehat{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\Vert_1$	57.147 (3.831)	28.204 (2.138)	16.051 (0.956)	69.757 (4.084)	42.229 (1.897)	30.434 (1.219)	97.181 (3.537)	70.548 (2.174)	58.815 (1.507)
$\mathbf{A}^*$ Sparsity	0.912 (0.003)	0.920 (0.003)	0.929 (0.002)	0.912 (0.003)	0.920 (0.003)	0.928 (0.003)	0.911 (0.003)	0.919 (0.003)	0.928 (0.003)
$\Vert\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\Vert_1$	0.055 (0.023)	0.041 (0.015)	0.026 (0.009)	0.054 (0.022)	0.041 (0.014)	0.025 (0.009)	0.058 (0.020)	0.050 (0.015)	0.028 (0.008)
$\boldsymbol{\delta}^*$ Specificity	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)	0.00% (0.000)
$\boldsymbol{\delta}^*$ Sensitivity	−% (−)	−% (−)	−% (−)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)	100.00% (0.000)
$\Vert\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}^*\Vert_1$	0.078 (0.025)	0.073 (0.025)	0.072 (0.023)	0.077 (0.021)	0.078 (0.025)	0.081 (0.022)	0.086 (0.028)	0.082 (0.027)	0.079 (0.023)
$\Vert\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}^*\Vert_{\max}$	0.310 (0.049)	0.176 (0.024)	0.125 (0.021)	0.303 (0.042)	0.174 (0.024)	0.119 (0.017)	0.307 (0.055)	0.191 (0.027)	0.123 (0.016)

Table 2: Penalizing $\boldsymbol{\delta}$

	$\boldsymbol{\delta}^* = \mathbf{0}$			$\boldsymbol{\delta}^* = 0.2\mathbf{e}_1$			$\boldsymbol{\delta}^* = 0.3(\mathbf{e}_1 + \mathbf{e}_2)$		
	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$
	$N = 25$								
$\mathbf{A}^*$ Specificity	96.06% (0.955)	96.69% (0.877)	97.09% (0.938)	95.94% (0.884)	96.64% (0.842)	97.02% (0.898)	95.59% (0.971)	96.30% (1.021)	96.48% (1.267)
$\mathbf{A}^*$ Sensitivity	85.05% (7.404)	98.35% (2.598)	99.97% (0.322)	83.37% (7.631)	98.40% (2.514)	100.00% (0.000)	82.86% (7.955)	98.16% (2.710)	99.97% (0.311)
$\left\Vert\widetilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\right\Vert_1$	19.020 (1.341)	17.356 (1.110)	14.452 (1.082)	22.282 (1.391)	20.472 (1.251)	17.408 (1.186)	28.904 (1.709)	27.216 (1.795)	25.253 (1.724)
$\left\Vert\widehat{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\right\Vert_1$	5.740 (0.856)	3.360 (0.583)	2.127 (0.406)	10.482 (0.845)	8.075 (0.566)	6.848 (0.393)	19.857 (1.019)	17.651 (0.800)	16.761 (0.680)
$\mathbf{A}^*$ Sparsity	0.921 (0.010)	0.921 (0.009)	0.924 (0.009)	0.921 (0.009)	0.920 (0.008)	0.923 (0.009)	0.917 (0.010)	0.917 (0.010)	0.918 (0.012)
$\left\Vert\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\right\Vert_1$	0.042 (0.023)	0.027 (0.015)	0.018 (0.010)	0.043 (0.022)	0.029 (0.016)	0.018 (0.010)	0.047 (0.025)	0.031 (0.016)	0.023 (0.012)
$\boldsymbol{\delta}^*$ Specificity	96.88% (7.710)	94.45% (9.690)	88.98% (13.790)	95.43% (10.318)	92.11% (13.421)	86.76% (16.522)	88.34% (18.474)	84.91% (20.276)	77.99% (22.197)
$\boldsymbol{\delta}^*$ Sensitivity	−% (−)	−% (−)	−% (−)	72.93% (44.470)	80.72% (39.486)	92.73% (25.981)	96.12% (13.690)	99.75% (3.553)	99.91% (2.065)
$\left\Vert\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}^*\right\Vert_1$	0.002 (0.007)	0.004 (0.008)	0.008 (0.012)	0.025 (0.017)	0.024 (0.017)	0.022 (0.015)	0.042 (0.027)	0.033 (0.021)	0.031 (0.018)
$\left\Vert\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}^*\right\Vert_{\max}$	0.186 (0.038)	0.119 (0.023)	0.082 (0.017)	0.197 (0.042)	0.128 (0.027)	0.090 (0.019)	0.198 (0.043)	0.137 (0.033)	0.092 (0.022)
	$N = 50$								
$\mathbf{A}^*$ Specificity	95.19% (0.517)	96.36% (0.464)	97.10% (0.465)	95.18% (0.489)	96.36% (0.415)	97.12% (0.416)	94.95% (0.535)	96.06% (0.592)	96.82% (0.690)
$\mathbf{A}^*$ Sensitivity	86.64% (3.995)	98.93% (1.061)	100.00% (0.035)	86.67% (3.943)	98.99% (1.058)	100.00% (0.000)	86.17% (3.868)	98.80% (1.176)	99.99% (0.084)
$\left\Vert\widetilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\right\Vert_1$	65.900 (2.617)	60.587 (2.269)	54.041 (1.872)	72.546 (2.671)	66.682 (2.151)	60.003 (1.974)	85.929 (2.820)	79.797 (2.895)	73.193 (3.082)
$\left\Vert\widehat{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\right\Vert_1$	23.940 (2.131)	13.183 (1.170)	8.000 (0.764)	33.125 (2.195)	22.438 (1.131)	17.327 (0.762)	51.860 (2.235)	41.742 (1.787)	36.971 (1.543)
$\mathbf{A}^*$ Sparsity	0.912 (0.005)	0.917 (0.004)	0.924 (0.004)	0.912 (0.005)	0.917 (0.004)	0.924 (0.004)	0.910 (0.005)	0.915 (0.006)	0.921 (0.007)
$\left\Vert\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\right\Vert_1$	0.043 (0.021)	0.028 (0.014)	0.017 (0.009)	0.044 (0.021)	0.029 (0.014)	0.018 (0.009)	0.047 (0.022)	0.034 (0.015)	0.021 (0.010)
$\boldsymbol{\delta}^*$ Specificity	95.64% (8.810)	93.25% (11.073)	86.54% (14.081)	93.36% (11.950)	89.32% (14.599)	83.07% (17.531)	80.66% (21.887)	75.92% (23.855)	70.11% (25.392)
$\boldsymbol{\delta}^*$ Sensitivity	−% (−)	−% (−)	−% (−)	54.13% (49.871)	71.12% (45.357)	83.77% (36.905)	91.64% (18.674)	97.19% (11.525)	99.92% (2.041)
$\left\Vert\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}^*\right\Vert_1$	0.004 (0.010)	0.006 (0.011)	0.012 (0.014)	0.032 (0.019)	0.030 (0.019)	0.029 (0.021)	0.057 (0.034)	0.051 (0.030)	0.045 (0.023)
$\left\Vert\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}^*\right\Vert_{\max}$	0.241 (0.039)	0.151 (0.023)	0.101 (0.014)	0.250 (0.039)	0.153 (0.023)	0.101 (0.016)	0.256 (0.043)	0.162 (0.025)	0.107 (0.018)
	$N = 75$								
$\mathbf{A}^*$ Specificity	95.20% (0.330)	96.69% (0.298)	97.69% (0.287)	95.15% (0.334)	96.65% (0.300)	97.65% (0.290)	95.00% (0.365)	96.42% (0.445)	97.19% (0.635)
$\mathbf{A}^*$ Sensitivity	85.17% (3.064)	98.81% (0.771)	100.00% (0.037)	84.91% (2.954)	98.79% (0.838)	99.99% (0.050)	84.33% (3.131)	98.71% (0.801)	99.99% (0.045)
$\left\Vert\widetilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\right\Vert_1$	131.119 (4.153)	117.364 (3.350)	106.129 (2.820)	141.602 (3.898)	127.437 (3.077)	115.824 (3.071)	163.206 (4.011)	148.457 (4.022)	137.031 (4.304)
$\left\Vert\widehat{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\right\Vert_1$	55.543 (3.750)	28.459 (2.031)	16.166 (1.142)	69.513 (3.725)	42.495 (2.142)	30.313 (1.195)	97.964 (3.907)	71.867 (3.243)	61.186 (3.206)
$\mathbf{A}^*$ Sparsity	0.913 (0.003)	0.920 (0.003)	0.929 (0.003)	0.912 (0.003)	0.920 (0.003)	0.929 (0.003)	0.911 (0.004)	0.918 (0.004)	0.924 (0.006)
$\left\Vert\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\right\Vert_1$	0.054 (0.023)	0.043 (0.015)	0.024 (0.008)	0.055 (0.023)	0.043 (0.014)	0.025 (0.009)	0.060 (0.022)	0.050 (0.015)	0.030 (0.010)
$\boldsymbol{\delta}^*$ Specificity	95.08% (9.576)	91.14% (12.294)	84.91% (14.453)	89.93% (14.758)	84.46% (16.632)	78.18% (20.172)	75.25% (25.020)	65.61% (27.570)	51.18% (26.955)
$\boldsymbol{\delta}^*$ Sensitivity	−% (−)	−% (−)	−% (−)	46.31% (49.905)	65.17% (47.684)	77.97% (41.483)	84.20% (24.833)	91.97% (18.370)	98.65% (8.102)
$\left\Vert\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}^*\right\Vert_1$	0.005 (0.011)	0.008 (0.013)	0.015 (0.017)	0.038 (0.021)	0.038 (0.024)	0.038 (0.026)	0.073 (0.043)	0.070 (0.041)	0.071 (0.036)
$\left\Vert\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}^*\right\Vert_{\max}$	0.299 (0.040)	0.176 (0.023)	0.112 (0.016)	0.309 (0.046)	0.181 (0.025)	0.116 (0.016)	0.316 (0.045)	0.187 (0.027)	0.127 (0.023)

Table 3: Variance of normalized $\widehat{\xi}_{J_1}$ and $\widehat{\boldsymbol{\beta}}$, as in equations (5.1) and (5.2)

	$\boldsymbol{\delta}^* = \mathbf{0}$			$\boldsymbol{\delta}^* = 0.2\mathbf{e}_1$			$\boldsymbol{\delta}^* = 0.3(\mathbf{e}_1 + \mathbf{e}_2)$		
	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$
					$N = 25$				
$V(\widehat{\boldsymbol{\xi}}_{J_1})$	1.799	1.286	0.996	1.866	1.349	0.991	1.906	1.443	1.114
$V(\widehat{\boldsymbol{\beta}})$	1.638	1.329	1.232	1.589	1.148	1.223	1.640	1.462	1.346
					$N = 50$				
$V(\widehat{\boldsymbol{\xi}}_{J_1})$	2.181	1.523	0.963	2.397	1.543	0.973	2.402	1.920	1.117
$V(\widehat{\boldsymbol{\beta}})$	2.029	1.648	1.448	2.189	1.736	1.570	2.119	1.770	1.469
					$N = 75$				
$V(\widehat{\boldsymbol{\xi}}_{J_1})$	1.991	1.589	0.937	2.355	1.561	0.963	2.978	1.808	1.182
$V(\widehat{\boldsymbol{\beta}})$	2.688	2.099	1.551	2.697	1.930	1.619	2.425	2.115	1.839

stationary series. Candidate $\mathbf{W}^*$ are the inverse of physical distances, measured from capital to capital, and the inverse squared in order to capture possible non-linearities. The physical distance is a measure that accounts for transportation costs[3] and possibly unobserved factors such as innovation and migration to some extent. We control of trade using bilateral export and import data, obtained from IMF's Direction of Trade Statistics.[4] Finally, we compile commonality of language and whether pairs of countries belong to common monetary union (in our sample, only the Eurozone).

To control for exogenous variation that are not due to spillover effects, we use the world GDP, total world exports, non-fuel commodity prices and oil prices.[5] All regressions include country-specific fixed-effects.

In Table 4 we report the results of four regressions, stacked along columns (1)-(4). For each regression we include a different setting of expert spatial distances. The results indicate that inverse of squared physical distance and monetary union are the major channels of GDP spillovers. Given that Eurozone membership is also closely associated to small distances, the results overall indicate that, when controlling for world GDP, geographic adjacency is the most important factor in explaining country-to-country GDP propagation.[6] In such cases, imports, exports and investments have no significant effect. This result highlights, for example, the potential large effect dissemination of a economic crisis in a Eurozone country. In all specifications, $\widehat{\mathbf{A}} = 0$.

Figure 1 presents the estimated combination of expert matrices and sparse deviations, that is, $\widehat{\mathbf{W}} = \widehat{\mathbf{A}} + \sum_{i=1}^{m} \widehat{\delta}_i \mathbf{W}_{0,i}$. The strength of GDP spillovers for country pairs, such as United States-Mexico, Japan-Korea and Germany-Netherlands, and country groups, such as Germany-England-France, are clearly distinguishable. Other countries, such as Brazil and Australia are economically isolated.

[3]Including iceberg costs, or product losses during transportation

[4]Taken at 2004 values. Adopting bilateral data at other years does not substantially change the results

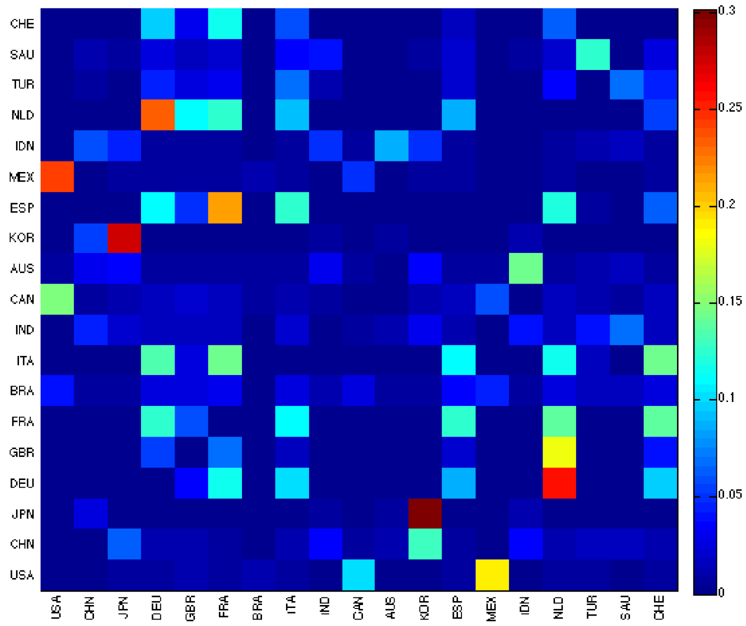
[5]Data source is the IMF's Primary Commodity Index.

[6]World GDP in most cases still has a economically significant effect.

Table 4: Regression results

		(1)	(2)	(3)	(4)
expert W weights	distance	.000	—	—	.000
	distance^2	.377	—	—	.390
	exports	—	.000	—	.000
	imports	—	.000	—	.000
	investment	—	.613	—	.000
	language	—	—	.000	.000
	monetary union	—	—	.875	.315
exog. vars.	world GDP	.346 (.345)	.119 (.499)	.541 (.368)	.633 (.331)
	total exports	.040 (.120)	.068 (.175)	.067 (.128)	.031 (.155)
	commodity price	.003 (.067)	−.001 (.097)	.004 (.066)	.001 (.086)
	oil price	−.006 (.097)	.013 (.141)	−.010 (.096)	.004 (.124)
	N	19	19	19	19
	T	35	35	35	35
	$\widehat{\gamma}_{\boldsymbol{\xi},T}$	.1	.1	.1	.1
	$\widehat{\gamma}_{\boldsymbol{\delta},T}$	5.653	5.584	0.001	5.576

Figure 1: Estimated $\widehat{W}$



Note: country codes are United States (USA), China (CHN), Japan (JPN), Germany (DEU), United Kingdom (GBR), France (FRA), Brazil (BRA), Italy (ITA), India (IND), Canada (CAN), Australia (AUS), South Korea (KOR), Spain (ESP), Mexico (MEX), Indonesia (IDN), Netherlands (NLD), Turkey (TUR), Saudi Arabia (SAU) and Switzerland (CHE). Rows represent equations and columns parameters in those equations. For example, the United States has a larger effect on Mexico than vice-versa.

Supplementary Material

All the proofs in the paper are presented in the supplementary materials in this paper.

6 Appendix

In this section, we present the regularity conditions of the paper which are more technically involved, and are not mentioned in the main body of the paper.

R1. Each column vector $\mathrm{vec}(\mathbf{W}_{0i}^{\mathrm{T}})$ in $\mathbf{V}_0$, $i = 1, \ldots, M$ is linearly independent of each other, such that there exists a constant $u > 0$ with $\sigma_M^2(\mathbf{V}_0) \geq u > 0$ uniformly as $N \to \infty$, where $\sigma_i(A)$ is the ith largest singular value of a matrix A.

Moreover, $\max_{1 \leq i \leq M} \|\mathbf{W}_{0i}\|_1 \leq c < \infty$ uniformly as $N \to \infty$ for some constant $c > 0$.

R2. Write $\boldsymbol{\epsilon}_t = \boldsymbol{\Sigma}_\epsilon^{1/2} \boldsymbol{\epsilon}_t^*$, where $\boldsymbol{\Sigma}_\epsilon$ is the covariance matrix for $\boldsymbol{\epsilon}_t$. Then the elements in $\boldsymbol{\Sigma}_\epsilon$ are all less than $\sigma_{\max}^2$ uniformly as $N \to \infty$. Same for the variance of the elements in $\mathbf{B}_t$.

We also assume $\|\boldsymbol{\Sigma}_\epsilon^{1/2}\|_\infty \leq S_\epsilon < \infty$ uniformly as $N \to \infty$, with $\{\epsilon_{t,j}^*\}_{1 \leq j \leq N}$ being a martingale difference with respect to the filtration generated by $\sigma(\epsilon_{t,1}^*, \ldots, \epsilon_{t,j}^*)$. The tail condition $P(|Z| > v) \leq D_1 \exp(-D_2 v^q)$ is also satisfied by $\epsilon_{t,j}^*$, while Assumption M4 is also satisfied by $\{\boldsymbol{\epsilon}_t^*\}_{1 \leq t \leq T}$.

R3. All singular values of $E(\mathbf{X}_t^{\mathrm{T}} \mathbf{B}_t)$ are uniformly larger than Nu for some constant $u > 0$, while the maximum singular value is also of order N. Individual entries in the matrix $E(\mathbf{b}_t \mathbf{x}_t^{\mathrm{T}})$ are uniformly bounded away from infinity, where $\mathbf{b}_t$ and $\mathbf{x}_t$ are defined in the paragraph containing (3.1).

R4. For $a \in [0, 1]$, define

$$\mathbf{G} = N^{-a} E(T^{-1} \mathbf{Z}^{\mathrm{T}} (\mathbf{B}_{\boldsymbol{\gamma}} - \bar{\mathbf{B}}_{\boldsymbol{\gamma}})) E(T^{-1} (\mathbf{B}_{\boldsymbol{\gamma}} - \bar{\mathbf{B}}_{\boldsymbol{\gamma}})^{\mathrm{T}} \mathbf{Z}).$$

Each block $E(T^{-1} \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \mathbf{y}_t^{\mathrm{T}}) = E(T^{-1} \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \boldsymbol{\beta}^{*\mathrm{T}} \mathbf{X}_t^{\mathrm{T}} \boldsymbol{\Pi}^{*\mathrm{T}})$ in the block diagonal matrix $E(T^{-1} (\mathbf{B}_{\boldsymbol{\gamma}} - \bar{\mathbf{B}}_{\boldsymbol{\gamma}})^{\mathrm{T}} \mathbf{Z})$ is assumed to have full rank N, such that there exists a constant $u > 0$ with $\lambda_{\min}(\mathbf{G}) \geq u > 0$ uniformly as $N \to \infty$. The maximum eigenvalue of $\mathbf{G}$ is uniformly bounded from infinity.

R5. For the same constant a in Assumption R4, we have for each N,

$$\max_{1 \leq i \leq N} \sum_{j=1}^N \|E(\mathbf{b}_{t,i} \mathbf{x}_{t,j}^{\mathrm{T}})\|_{\max}, \quad \max_{1 \leq j \leq N} \sum_{i=1}^N \|E(\mathbf{b}_{t,i} \mathbf{x}_{t,j}^{\mathrm{T}})\|_{\max} \leq C_{bx} N^a,$$

where $C_{bx} > 0$ is a constant and $\mathbf{b}_{t,i}$, $\mathbf{x}_{t,j}$ are the column vectors for the ith row of $\mathbf{B}_t$ and jth row of $\mathbf{X}_t$ respectively. At the same time, assume also that $E(\mathbf{X}_t \otimes \mathbf{B}_t\boldsymbol{\gamma})$ has all singular values of order N^{1+a}.

R6. Assume $0 < b < 1$. For fixed $1 \le k \le K$, the eigenvalues of $\mathrm{var}(N^{-b/2}\mathbf{B}_{t,k})$ and $\mathrm{var}(\boldsymbol{\epsilon}_t)$ are uniformly bounded away from 0 and infinity, and respectively dominates the singular values from $N^{-b}E((\mathbf{B}_{t+\tau,k} - \boldsymbol{\mu}_{b,k})(\mathbf{B}_{t,k} - \boldsymbol{\mu}_{b,k})^{\mathrm{T}})$ and $E(\boldsymbol{\epsilon}_t\boldsymbol{\epsilon}_{t+\tau}^{\mathrm{T}})$. The sum of the ith largest singular values over τ for each $1 \le i \le N$ is assumed finite for both $\{N^{-b/2}\mathbf{B}_t\}$ and $\{\boldsymbol{\epsilon}_t\}$.

R7. Define $\lambda_T = cT^{-1/2}\log^{1/2}(T \vee N)$ for some constant $c > 0$. The tuning parameter γ_T for (2.8) is such that $\gamma_T = C\lambda_T$ for some constant $C > 0$. The tuning parameter γ_T' for (2.11) is $\gamma_T' = C'\lambda_T$ for some constant $C' > 0$.

R8. In all the assumptions above, we assume that as $N, T \to \infty$,

$$\lambda_T N^{1-a},\ N^{-1/2w}\log^{1/2}(T \vee N),\ \lambda_T n,\ nN^{a-1}\log^{1/2}(T \vee N),\ nN^{-\frac{1}{2}-\frac{1}{2w}} = o(1),$$
$$T^{-1/2}N^{\frac{a-b}{2}}\log(T \vee N),\ \log(T \vee N)N^{-b+\frac{1}{w}}, T^{-1/2}N^{\frac{b}{2}+\frac{1}{2w}} = o(1),$$

where $n = |J_1|$, with J_1 defined in (3.3). We further assume $T^{1/2}N^{\frac{a-b}{2}}\max_{1\le j\le N}\left\|\mathbf{a}_{j,J_2}^*\right\|_1 = o(1)$ and $\left\|\boldsymbol{\xi}_{J_2}^*\right\|_1 = o(\lambda_T N^{\frac{1}{2}+\frac{1}{2w}}) = o(1)$, where $\mathbf{a}_j^*$ is the column vector of the jth row of $\mathbf{A}^*$.

Assumption R1 essentially requires that each specification $\mathbf{W}_{0i}$ is different from each other to a certain extent. This is intuitive, since if $\mathbf{W}_{0i}$ and $\mathbf{W}_{0j}$ are too similar to each other, the coefficients δ_i^* and δ_j^* are not well defined, and this will have a negative impact on the performance of our estimators.

The assumptions on $\boldsymbol{\Sigma}_\epsilon$ in R2 is mainly for convenience of proof, while the martingale difference assumption for $\boldsymbol{\epsilon}_t^*$ is a relaxation to independence.

Assumption R3, R4 and R5 are closely related. They all paint a picture of how the exogenous variables in $\mathbf{B}_t$ are correlated with $\mathbf{X}_t$. Assumption R3 essentially says that covariance between a variable in $\mathbf{B}_t$ and one in $\mathbf{X}_t$ is finite uniformly as $N \to \infty$. Then for $1 \le k \le K$, considering the kth diagonal entry of $E(X_t^{\mathrm{T}}\mathbf{B}_t)$ to be $\sum_{j=1}^N E(X_{t,jk}B_{t,jk})$ with each $E(X_{t,jk}B_{t,jk})$ finite, it is indeed reasonable to assume that each diagonal entry in the matrix is of order N, so that it is also reasonable to assume that this finite $K \times K$ matrix has all singular values of order N. This assumption is needed for the estimator $\widetilde{\boldsymbol{\beta}} = \boldsymbol{\beta}(\widetilde{\boldsymbol{\theta}})$ to be well-defined in (2.8).

Assumption R4 is closely related to the identification condition M5, since the upper left block of $\mathbf{Q}^{\mathrm{T}}\mathbf{Q}$ is in fact $\mathbf{G}_{SS}$, and the identification condition already implies that $\mathbf{G}_{SS}$ has all eigenvalues uniformly bounded away from 0. Assumption R4 places a bit more stringent condition in the sense that the whole matrix $\mathbf{G}$ is now of full rank so that all eigenvalues of $\mathbf{G}$ are also uniformly bounded away from 0 and infinity. We do so because it can avoid the need for a restricted eigenvalue condition which is technically very involved and unnecessarily complicated in our context.

Assumption R5 essentially describes how each row of variables in $\mathbf{B}_t$ are correlated with different rows of variables in $\mathbf{X}_t$. The rate N^a is closely related to Assumption R4. This can be seen by noting that each block $E(T^{-1}\sum_{t=1}^T(\mathbf{B}_t - \bar{\mathbf{B}})\boldsymbol{\gamma}\mathbf{y}_t^{\mathrm{T}})$ in $\mathbf{G}$ is asymptotically the same as $(I_N \otimes \boldsymbol{\gamma}^{\mathrm{T}})\mathbf{L}_0(\mathbf{I}_N \otimes \boldsymbol{\beta}^*)\boldsymbol{\Pi}^{*\mathrm{T}}$, where $\mathbf{L}_0 = E(\mathrm{vec}(\mathbf{B}_t^{\mathrm{T}})\mathrm{vec}(\mathbf{X}_t^{\mathrm{T}})^{\mathrm{T}})$. Note that the (i,j)th $K \times K$ block in $\mathbf{L}_0$ is indeed $E(\mathbf{b}_{t,i}\mathbf{x}_{t,j}^{\mathrm{T}})$. Assumption R4 then essentially says that both

$$\max_{1\leq i\leq N}\sum_{j=1}^N \left\|E(\mathbf{b}_{t,i}\mathbf{x}_{t,j}^{\mathrm{T}})\right\|_{\max}^2, \quad \max_{1\leq j\leq N}\sum_{i=1}^N \left\|E(\mathbf{b}_{t,i}\mathbf{x}_{t,j}^{\mathrm{T}})\right\|_{\max}^2$$

are of order similar to N^a. Assumption R5 can then be seen as the regularity condition such that the upper bounds (with square removed for ease of proofs) are exactly of order N^a. With this, we can actually derive easily that $\left\|E(\mathbf{X}_t \otimes \mathbf{B}_t\boldsymbol{\gamma})\right\|_1$ has order at most N^{1+a}. Hence the assumption of having all the singular values of $E(\mathbf{X}_t \otimes \mathbf{B}_t\boldsymbol{\gamma})$ of order N^{1+a} is reasonable.

Assumption R6 assumes a rate for the singular values of $\mathrm{var}(\mathbf{B}_{t,k})$ essentially, which is important in certain asymptotic normality results. The rate N^b, possibly differing from N^a, is reasonable as well since the way that $\mathbf{B}_t$ and $\mathbf{X}_t$ are correlated do not directly indicate how the variables in $\mathbf{B}_t$ itself are correlated, unless of course when $\mathbf{B}_t = \mathbf{X}_t$ when $\mathbf{X}_t$ itself is exogenous, in which case $b = a$. The variance-covariance matrix being dominating the lag-τ auto-covariances is for the ease of presentation of rates of convergence in asymptotic normality results in this paper.

With the rates assumed in R8, the rates of the tuning parameters γ_T and γ_T' for the penalization problems (2.8) and (2.11) respectively can be set the same as λ_T, which is Assumption 7.

References

Ahrens, A. and A. Bhattacharjee (2015). Two-step lasso estimation of the spatial weights matrix. *Econometrics 3*(1), 128.

Ammermuller, A. and J.-S. Pischke (2009). Peer effects in european primary schools: Evidence from pirls. *Journal of Labor Economics 27*(3), 315–348.

Angrist, J. D. and K. Lang (2004, December). Does school integration generate peer effects? evidence from boston's metco program. *American Economic Review 94*(5), 1613–1634.

Arnold, M., S. Stahlberg, and D. Wied (2011). Modeling different kinds of spatial dependence in stock returns. *Empirical Economics 44*(2), 761–774.

Beenstock, M. and D. Felsenstein (2012). Nonparametric estimation of the spatial connectivity matrix using spatial panel data. *Geographical Analysis 44*(4), 386–397.

Bhattacharjee, A. and C. Jensen-Butler (2013). Estimation of the spatial weights matrix under structural constraints. *Regional Science and Urban Economics 43*(4), 617 – 634.

Chen, X., M. Xu, and W. B. Wu (2013, 12). Covariance and precision matrix estimation for high-dimensional time series. *The Annals of Statistics 41*(6), 2994–3021.

Corrado, L. and B. Fingleton (2011, Jan). Where is the economics in spatial econometrics? Working Papers 1101, University of Strathclyde Business School, Department of Economics.

Dicker, L., B. Huang, and X. Lin (2013). Variable selection and estimation with the seamless-l0 penalty. *Statistica Sinica 23*, 929–962.

Fan, J. and Y. Liao (2014, 06). Endogeneity in high dimensions. *Ann. Statist. 42*(3), 872–917.

Fan, J. and J. Lv (2011). Non-concave penalized likelihood with np-dimensionality. *IEEE Transactions on Information Theory 57*, 5467–5484.

Friedman, J., T. Hastie, and R. Tibshirani (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software 33*, 1–22.

Glaeser, E., B. Sacerdote, and J. A. Scheinkman (1996, May). Crime and social interactions. *The Quarterly Journal of Economics 111*(2), 507–548.

Lam, C. and P. C. L. Souza (2015a). Detection and estimation of block structure in spatial weight matrix. *Econometric Reviews*, 1–30.

Lam, C. and P. C. L. Souza (2015b). One-step regularized spatial weight matrix and fixed effects estimation with instrumental variables. Manuscript.

Lee, L.-F. and J. Yu (2010). Estimation of spatial autoregressive panel data models with fixed effects. *Journal of Econometrics 154*, 165–185.

Longstaff, F. A. (2010, September). The subprime credit crisis and contagion in financial markets. *Journal of Financial Economics 97*(3), 436–450.

Pinkse, J., M. E. Slade, and C. Brett (2002). Spatial price competition: A semiparametric apprach. *Econometrica 70*(3), 1111–1153.

Shao, X. (2010). Nonstationary-extended whittle estimation. *Econometric Theory 26*, 1060–1087.

Wang, H., B. Li, and C. Leng (2009). Shrinkage tuning parameter selection with a diverging number of parameters. *Journal of the Royal Statistical Society: Series B (Statistical Methodology) 71*(3), 671–683.

Wu, W. B. (2005). Nonlinear system theory: another look at dependence. *Proc. Natl. Acad. Sci. USA 102*, 14150–14154.

Wu, W. B. (2011). Asymptotic theory for stationary processes. *Statistics and Its Interface 4*, 207–226.

Zhao, P. and B. Yu (2006). On model selection consistency of lasso. *Journal of Machine Learning Research 7*, 2541–2563.

Zhou, Z. (2010). Nonparametric inference of quantile curves for nonstationary time series. *Ann. Statist. 38*(4), 2187–2217.

Zou, H. (2006, December). The adaptive lasso and its oracle properties. *Journal of the American Statistical Association 101*, 1418–1429.